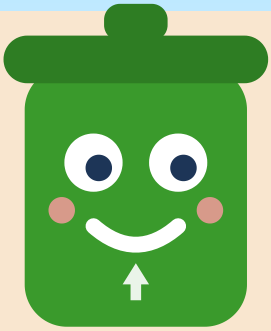
WasteQuest

Waste-to-Wealth (W2W) Educational Module

Interactive website · Games & virtual labs · 15 hands-on activities · Class quiz · Certificates



Open the website
wastequest.github.io/

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Facilitator booklet · 2026 edition

Contents

1. About WasteQuest
2. Learn: Waste-to-Wealth
3. Games & Simulations
4. Hands-on Labs
5. Teacher Guide & Certificates
6. Appendix: Question Bank & Answers

1. About WasteQuest

WasteQuest turns the journey of waste, from the bin to a valuable new product, into a quest. Learners play games, run virtual science labs, make real products from waste in hands-on activities, compete in live class quizzes and earn badges and a certificate.

It serves four groups: **children 7–12**, **teenagers 13–17**, **adults** (university, community and ESG training) and **teachers/facilitators**. Every page is available in English and Bahasa Melayu and works on phones, tablets and classroom projectors, online or as a single offline file.

How to use the website

Section	What learners do
📘 Learn	Seven short interactive chapters with quick checks.
🎮 Games & Simulations	Sorting, matching, puzzles, quizzes and five virtual labs. Each earns a badge.
🧪 Hands-on Labs	Fifteen step-by-step activities with materials, safety notes, science, videos and printable worksheets.
📱 Class Battle	The teacher hosts a live quiz on the projector; students join with a code on their phones. Team mode works without phones.
🏅 Certificate	Three tracks: Junior Eco-Hero, Zero-Waste Champion and Waste-to-Wealth Practitioner (non-credit Certificate of Completion).
👩‍🏫 Teacher Hub	Session plans, certificate design, rubrics and safety guidance.

2. Learn: Waste-to-Wealth

CHAPTER 1

🗑️ Where does our waste go?

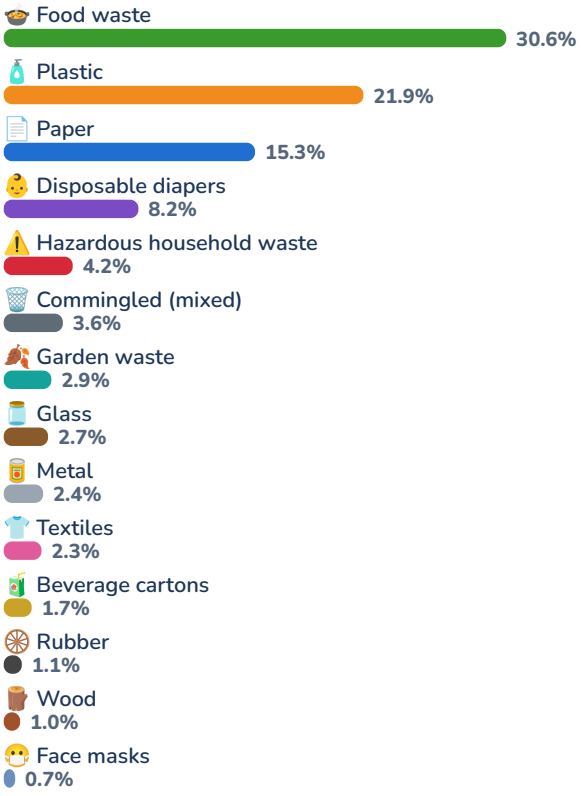
Malaysia generates about 39,078 tonnes of solid waste every day (SWCorp, 2024). Most of it still ends up in landfills, where it takes space, pollutes and releases greenhouse gases.

39,078 t
of solid waste per day in Malaysia (2024)

1.17 kg
per person, per day

×2
daily waste roughly doubled from 2005 to 2024

What is in a Malaysian household bin?



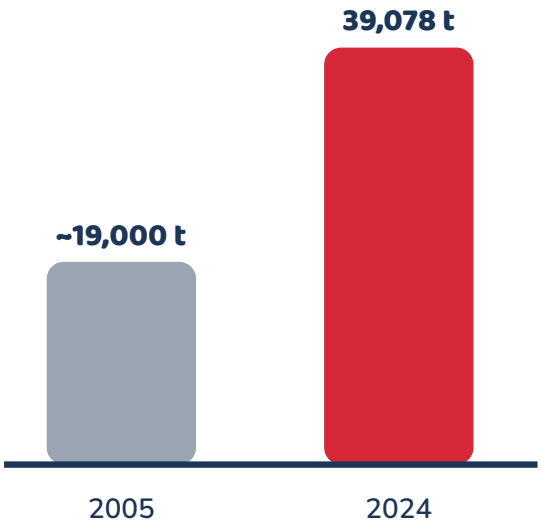
Household waste composition, % by weight. Source: SWCorp via The Star, 2 Jan 2024.

Your 1.17 kg a day, stacked up



Rough split using the household mix above.

Daily waste: 2005 vs 2024



💡 Food (30.6%), plastic (21.9%) and paper (15.3%) make up about two-thirds of household waste, and all three can be composted or recycled.

🔬 The science

In a landfill, buried food waste has no oxygen. Anaerobic microbes break it down into **landfill gas**, roughly 50% methane (CH_4) and 50% carbon dioxide (US EPA). For sugar, a simplified equation is **$C_6H_{12}O_6 \rightarrow 3\ CH_4 + 3\ CO_2$**.

Methane traps far more heat than CO_2: its global warming potential over 100 years is about **28** (IPCC AR5, 2013). So 1 kg of methane ≈ 28 kg CO_2-equivalent. Composting with air (aerobic) gives mainly CO_2 and water instead.

Maths: 1.17 kg × 365 days ≈ **427 kg** per person per year.

💼 ESG and business angle

Malaysia's national recycling rate was reported as 37.9% in 2024 (KPKT, via Bernama 2025); experts note this counts recyclables collected, not necessarily recycled. For organisations, waste is a standard ESG disclosure topic (e.g. GRI 306: Waste 2020): tonnes generated, diverted from disposal and sent to disposal.

👩‍🏫 Teaching note

15 min. Ask: “What did you throw away today?” before showing the chart. Let groups guess the top 3 materials, then reveal. Follow with Sort It Out (#/game/sort).

✅ Quick check

1. Why is food waste in a landfill a climate problem?

a. It decomposes without oxygen and releases methane
b. It turns into plastic
c. It absorbs carbon dioxide

Answer: a. Anaerobic decomposition releases methane, with a 100-year GWP of about 28 (IPCC AR5).

2. Daily solid waste in Malaysia went from ~19,000 t (2005) to ~39,000 t (2024). That is an increase of about…

a. 20%
b. 50%
c. 100% (it doubled)

Answer: c. (39,078 – 19,000) ÷ 19,000 ≈ 1.06, so just over 100%.

CHAPTER 2

🔺 The waste hierarchy

The waste hierarchy ranks options from most to least preferred. The higher the step, the more value we keep and the less energy and raw material we use.



1. 🙅 **Refuse:** Stop waste before it exists: turn down single-use items and freebies you will not use. *Plastic straws, extra plastic bags, flyers, free pens you will never use.*
2. 📉 **Reduce:** Use fewer resources: buy less, choose less packaging and plan meals so food is not wasted. *Refill bottles, print double-sided, buy in bulk, take smaller food portions.*
3. 🔁 **Reuse:** Keep products in use for longer: repair, share, donate or repurpose them (upcycling). *Old T-shirt to tote bag, bottle to planter, jam jar to pencil holder.*
4. ♻️ **Recycle:** Turn waste materials back into raw materials: paper, plastic, metal and glass go to the SWCorp bins; food waste can be composted or fermented. *Paper to new paper, cans to new cans, peels to compost or eco-enzyme, used cooking oil to candles.*
5. ⚡ **Recover:** Recover energy from what is left, for example biogas from food waste or electricity from waste-to-energy plants. *Biogas digesters, waste-to-energy plants.*
6. 🚮 **Dispose:** Safe disposal in a sanitary landfill is the last resort: all the value in the material is lost. *Used tissues, dirty masks, broken mirrors (wrapped safely).*

🔬 **The science**

Why is higher better? Each step down loses more of the energy and materials already invested. Example: recycling aluminium cans saves more than 90% of the energy needed to make new aluminium from bauxite ore (The Aluminum Association), but refusing the can saves 100%.

“Refuse” comes from the zero-waste movement (Bea Johnson, 2013); official hierarchies such as the EU Waste Framework Directive start at prevention.

> **💼 ESG and business angle**
>
> Business translation: Refuse/Reduce = redesign packaging and processes; Reuse = refill and take-back schemes, repair services; Recycle = closed-loop material contracts; Recover = biogas or energy from residues. Moving up the hierarchy usually cuts disposal fees as well as emissions.

> **👩‍🏫 Teaching note**
>
> 10 min. Give each group a waste item card and ask them to place it on the pyramid. Discuss: why is recycling not the top?

✅ Quick check

1. Put in order, best first: Recycle, Reduce, Dispose

a. Recycle → Reduce → Dispose
b. Reduce → Recycle → Dispose
c. Dispose → Recycle → Reduce

Answer: b. Reducing avoids waste; recycling recovers material; disposal loses it.

2. Turning an old T-shirt into a tote bag is an example of...

a. Reuse (upcycling)
b. Recover
c. Dispose

Answer: a. The product gets a new job without being broken down into raw material.

CHAPTER 3

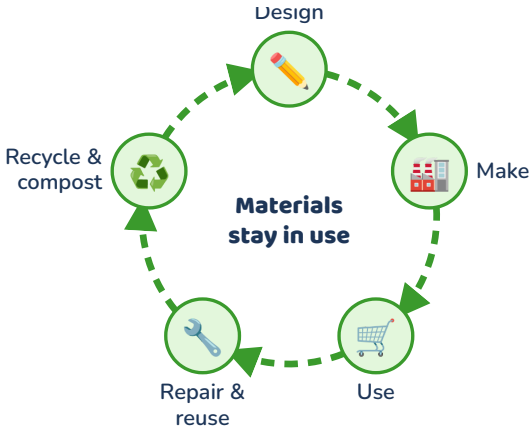
🔄 Linear vs circular economy

A linear economy follows “take, make, use, throw away”. A circular economy designs waste out, keeps products and materials in use at their highest value, and regenerates nature (Ellen MacArthur Foundation).

Linear



Circular



	➡️ Linear	🔄 Circular
Design	Cheap, single-use, hard to repair	Durable, repairable, easy to take apart and recycle
End of life	Landfill or open burning	Becomes a resource for a new product, or compost
Example	Used cooking oil poured down the sink	Used cooking oil made into candles (🕯️ **Eco-Candles from Used Cooking Oil**)

🔬 **The science**

Circular thinking uses two loops. The **biological cycle**: food and plant materials go back to the soil safely (composting, eco-enzyme). The **technical cycle**: metals, plastics and glass are kept in use by reuse, repair, remanufacture and recycling. Problems start when the two are mixed, such as food stuck to plastic.

💼 **ESG and business angle**

Circular business models: product-as-a-service (leasing), refill and reuse systems, take-back and remanufacture, and selling by-products. **Extended Producer Responsibility (EPR)** makes producers responsible for their packaging after use. In Malaysia, KPKT has announced a voluntary EPR phase for packaging from 2026, becoming mandatory by 2030 (Bernama, 2025).

> 🧑‍🏫 **Teaching note**
> 10 min. Toggle the diagram on the projector. Ask learners to trace one product (a drink bottle) through both systems.

✅ Quick check

1. Extended Producer Responsibility (EPR) means…

a. Consumers pay all recycling costs
b. Producers take responsibility for their products and packaging after use
c. Local councils ban packaging

Answer: b. In Malaysia, a voluntary phase for packaging is planned from 2026, mandatory by 2030.

2. Which is a circular business model?

a. Refill and take-back scheme
b. Larger single-use packs
c. Planned obsolescence

Answer: a. Refill and take-back keep products and materials in use.

CHAPTER 4

💎 Waste-to-Wealth (W2W)

Waste-to-Wealth (W2W) turns waste into products, materials or energy with value. It is the circular economy in action: it reduces, reuses, recycles and recovers, and extends the life of products.

Three kinds of benefit

🌍 Environmental

- Less pollution of air, rivers and sea
- Lower greenhouse gas emissions (less methane from landfills)
- Conserves raw materials, water and energy

💰 Economic

- Value-added products made from cheap or free materials
- Green entrepreneurship and small businesses
- New jobs and income opportunities

🤝 Social

- Greater awareness in schools and communities
- Sustainable lifestyles and habits
- Innovation, creativity and teamwork

From waste to wealth: our 18 labs

Used cooking oil
🕯️ **Eco-Candles from Used Cooking Oil** →

Fish waste
🐟 **Fish-Waste Pet Treats** →

Used oil, old T-shirts and bottle caps
🎁 **Trash to Treasure: Soap, Tote & Toy Car** →

Coloured plant scraps
🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste** →

Used coffee grounds
☕ **Kopi Wira Bau: Coffee-Ground Odour Absorbers** →

Plastic bottles
🌱 **Self-Watering Bottle Garden** →

Fruit and vegetable peels
🍊 **Eco-Enzyme** →

Soft plastic wrappers
🧱 **Eco-Bricks & Eco-Bags** →

Food and garden waste
🪱 **Composting: Bokashi & Worms** →

Plant starch (instead of plastic)
🌽 **Cornstarch Bioplastic** →

Plastic bottles and compost
🪴 **Bottle Vertical Garden** →

Plastic bottles
🥬 **Bottle Hydroponics** →

Plastic bags and straws
☂️ **Fused-Plastic Bags & Mini Umbrella** →

Bubble wrap
🛟 **Bubble-Wrap Float Ring (Model)** →

Bubble wrap and old cloth

🛌 **3-in-1 Bubble-Wrap Sleeping Bag** ➜

🔬 The science

Behind each product is real science: microbes turn food waste into compost (compost sim); fermentation makes eco-enzyme acidic (eco-enzyme sim); plant pigments called anthocyanins change colour with pH (pH lab); glycerol acts as a plasticiser to make starch bioplastic flexible.

💼 ESG and business angle

Green entrepreneurship starts with cheap inputs. UPM ENG3104 student teams (2024) estimated a starch bioplastic pot at about RM0.84 in materials and a tote bag from used plastic bags at about RM4.66 (student estimates, not market prices). Test your own pricing in Waste to Cash. A credible W2W business also needs a safe process, a steady waste supply, quality control and honest environmental claims.

👩‍🏫 Teaching note

15 min. Flip the three cards with the class, then let each group choose the lab they will do in the hands-on session.

✅ Quick check

1. Which is a SOCIAL benefit of W2W?

a. Lower greenhouse gas emissions
b. Greater community awareness and new skills
c. Higher product prices

Answer: b. Awareness, lifestyles and creativity are social benefits.

2. Fruit peels + sugar + water, fermented, make...

a. Eco-enzyme
b. Bioplastic
c. Candles

Answer: a. Microbes ferment the sugars and the liquid becomes acidic.

CHAPTER 5

🏛️ Malaysia's rules and bins

Under the Solid Waste and Public Cleansing Management Act 2007 (Act 672), separation at source is mandatory in the states and territories that adopted the Act. Recyclables are separated from other household waste for collection.

Where separation at source is mandatory (Act 672):

Johor · Melaka · Negeri Sembilan · Pahang · Perlis · Kedah · Kuala Lumpur · Putrajaya

Other states run their own programmes. Selangor is working towards adopting the Act in phases; the start date is not yet confirmed, so check your council's rules.

📄 Blue bin: paper

✅ Newspapers, magazines, cardboard boxes (flattened), exercise books, envelopes, paper bags.

❌ Used tissues, oily pizza boxes, wet or food-soiled paper.

🥫 Orange bin: plastic & aluminium/metal

✅ Rinsed plastic bottles and containers, aluminium drink cans, food tins, clean plastic bags.

❌ Oily food boxes, used face masks, styrofoam with food left in it.

🫙 Brown bin: glass

✅ Glass bottles and jars (rinsed, lids removed).

❌ Mirrors, window glass, light bulbs, ceramic plates. Wrap broken glass safely.

3 easy steps

1️⃣ Rinse · 2️⃣ Dry · 3️⃣ Separate into the right bin. Food waste and dirty items go with general waste, or better, into compost. Batteries, e-waste and medicine go to special collection points.

🔬 The science

Why rinse? Food and oil left on packaging contaminate other recyclables: wet, greasy paper fibres cannot be pulped well, and dirty plastic lowers the quality of recycled pellets. Clean, dry, separated materials are worth more to recyclers.

💼 ESG and business angle

Policy context for organisations: Act 672 (separation at source), Malaysia's Roadmap Towards Zero Single-Use Plastics 2018–2030, and the planned EPR scheme for packaging (voluntary from 2026, mandatory by 2030). Workplaces can mirror the household system with labelled blue, orange and brown bins plus a food-waste stream.

👩‍🏫 Teaching note

10 min. Bring three coloured boxes and 10 real (clean) items. Then play Sort It Out on the projector, Level 1 then Level 2.

✅ Quick check

1. Which law makes separation at source mandatory in adopting states?

a. Act 672 (2007)
b. Environmental Quality Act 1974
c. There is no such law

Answer: a. The Solid Waste and Public Cleansing Management Act 2007 (Act 672).

2. An aluminium drink can goes into the…

a. Blue bin
b. Orange bin
c. Brown bin

Answer: b. Orange = plastic and aluminium/metal.

CHAPTER 6

🎯 The Sustainable Development Goals

In 2015, all UN member states, including Malaysia, adopted 17 Sustainable Development Goals (SDGs) for 2030. Waste-to-Wealth links most directly to SDGs 12, 13 and 15, and also to 11 and 14.



12 Responsible consumption and production

Target 12.5: substantially reduce waste generation through prevention, reduction, recycling and reuse by 2030. Target 12.3: halve per-capita food waste.

- 🪱 Composting: Bokashi & Worms
- 🍊 Eco-Enzyme
- 🎁 Trash to Treasure: Soap, Tote & Toy Car
- 🕯️ Eco-Candles from Used Cooking Oil
- 🐟 Fish-Waste Pet Treats
- ☕ Kopi Wira Bau: Coffee-Ground Odour Absorbers

13 Climate action

Keeping food waste out of landfills cuts methane, a strong greenhouse gas. Recycling saves the energy needed to make new materials.

- 🪱 Composting: Bokashi & Worms
- 🍊 Eco-Enzyme
- 🌽 Cornstarch Bioplastic

15 Life on land

Compost returns nutrients to soil; less landfill means less land cleared and less leachate polluting soil and groundwater.

- 🪱 Composting: Bokashi & Worms
- 🌱 Self-Watering Bottle Garden
- 🌈 Say No to Litmus: Rainbow pH from Kitchen Waste

11 Sustainable cities and communities

Target 11.6: reduce the environmental impact of cities, including better municipal waste management.

- 🧱 Eco-Bricks & Eco-Bags

14 Life below water

Target 14.1: reduce marine pollution, much of which comes from land, such as plastic litter.

- 🧱 Eco-Bricks & Eco-Bags
- 🎁 Trash to Treasure: Soap, Tote & Toy Car

💼 ESG and business angle

Organisations often map their ESG reports to the SDGs. A W2W project can report outputs (kg of waste diverted, products made) and link them to Target 12.5 (waste reduction) and Target 12.3 (food waste).

👩‍🏫 Teaching note

5–10 min. Ask each group to name which SDG their lab supports and why, then add it to their pitch poster.

✅ Quick check

1. Composting school food waste instead of landfilling it mainly supports...

a. SDG 12 and SDG 13
b. SDG 1 only
c. SDG 9 only

Answer: a. Less waste (12) and less methane (13), plus better soil (15).

2. SDG Target 12.3 aims to...

a. Halve per-capita food waste
b. Double plastic production
c. Build more landfills

Answer: a. Target 12.3 is about food loss and waste.

CHAPTER 7

🏆 Be a Zero-Waste Champion

Build a personal (or team) action plan. Pick specific, measurable actions, set a start date and review after 30 days.

📝 My Zero-Waste action plan

Name

Refuse

- [ ] Say no to plastic straws and extra bags
- [ ] Bring my own water bottle and food container

Reduce

- [ ] Take only the food I can finish
- [ ] Print on both sides of the paper

Reuse

- [ ] Use a reusable shopping bag
- [ ] Give old clothes, toys and books a second life

Recycle

- [ ] Rinse and sort recyclables into the blue, orange and brown bins
- [ ] Start a compost bin or an eco-enzyme bottle

Share

- [ ] Teach my family how to sort waste
- [ ] Join or start an Eco-Club

My own idea

I will start on

What does a Zero-Waste Champion do?
Practises the waste hierarchy daily, leads by example in school, measures progress (e.g. weighs the class bin each week) and shares ideas in the community.

👩‍🏫 **Teaching note**
10 min. Learners complete and print the plan as an exit ticket. Review the plans after 2–4 weeks (Eco-Club week 4).

✅ Quick check

1. Which goal is the most useful for an action plan?

a. “Be greener”
b. “Bring my own bottle every school day for 4 weeks”

Answer: b. Specific, measurable and time-bound goals are easier to keep and check.

2. How can your class measure its progress?

a. Weigh the general-waste bin every week
b. Guess

Answer: a. Data shows whether actions really reduce waste.

3. Games & Simulations

Scan a code to open that game directly.

Games

🗑️

Sort It Out!

Drag each item into the right Malaysian recycling bin. · 7+

🃏

Eco-Match

Flip the cards. Match each waste with the useful product it becomes. · 7+

✏️

Let's Cross Together!

Solve a waste-to-wealth crossword. A new puzzle every time. · 8+

🔍

Fun Puzzle: Word Search

Find the hidden waste-to-wealth words in the grid. · 7+

🕵️

Myth or Fact?

Swipe each card: is it a myth or a fact? Learn the real story behind each one. · 7+

⚡

Eco Quiz Blast

Beat the 20-second clock! Fast answers score more points. · 7+

Simulations

🧪

Say No to Litmus!

Test kitchen liquids with natural indicators from cabbage, bunga telang and kunyit. · 7+

🌱

Compost Master

Build a compost bin, balance greens and browns, and simulate 8 weeks of hot composting. · 7+

🍊

Eco-Enzyme Mixer

Solve the 1:3:10 recipe for your bottle, then fast-forward 3 months of fermentation. · 7+

💰

Trash-to-Cash Tycoon

Run a 10-day waste-to-wealth business: turn neighbourhood waste into products and profit. · 7+



My Waste Footprint

Count a week of your family's waste, compare with Malaysia, then make a pledge card. · 7+

4. Hands-on Labs

🕯️ Eco-Candles From Used Cooking Oil

Turn the oil left in the wok into a glowing scented candle.

Ages 10+ · ⏱ 60 min + 2–4 h to set · Difficulty ★★☆ · Adult does hot or sharp steps · 🔥 Needs heat · SDG 6 · SDG 12 · SDG 13

> 🧒 **Age and supervision:** Primary pupils (up to 12): an adult does all the heating and pouring as a demonstration. Children filter cold oil, weigh and decorate.

Waste used: Used cooking oil · **Product:** Scented candles · **Cost:** Low: the oil is free. Example shop prices (Oct 2026): paraffin wax RM 14–22 per kg, so the 96 g in this recipe is about RM 1.30–2.10; wicks about RM 0.50 each; fragrance extra (one 10 mL bottle was RM 25). Prices vary by shop.

Why it matters

🌍 **Environment:** Oil poured down the sink sticks inside pipes, causes blockages and pollutes rivers. Every jar of oil you reuse stays out of the drain.

💰 **Economy:** A free waste becomes a product people buy as gifts. Used cooking oil also has cash value: some recycling centres and collection points buy it for biodiesel.

🤝 **Society:** Candle-making is a simple home-business skill and a family activity that starts a conversation about how we throw away oil.



Steps poster (BM). Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- 48 g used cooking oil, cooled and left to settle
- 96 g paraffin wax (pellets or chips)
- Candle wicks with metal tabs, one per candle
- Moulds: small aluminium cups, clean tins or heat-proof glass jars
- Filter paper or a coffee filter, a funnel and a clean jar
- Kitchen scale
- Double boiler: a pot of water + a heat-proof jug or clean tin
- Wooden stick for stirring
- Candle dye or wax-crayon shavings (not food colouring)
- Candle fragrance oil or essential oil, a few drops
- Clothes pegs or chopsticks to hold the wicks straight
- Oven gloves, apron, a damp cloth and a pot lid close by

Steps

1. Ask an adult for cooled used cooking oil. Let it stand for a day so crumbs sink to the bottom.
2. Filter the oil through filter paper into a clean jar. Go slowly: it takes time!
   Food bits left in the oil do not burn cleanly: they char on the wick, smoke and smell.
3. Weigh 48 g of filtered oil and 96 g of paraffin wax. That is 1 part oil to 2 parts wax.
   Oil : wax = 48 : 96 = 1 : 2. Oil is liquid at room temperature, so too much oil gives a soft, greasy candle.
4. Stand a wick in the middle of each mould. Hold it straight with a clothes peg or two chopsticks across the top.
5. ADULT: put a few cm of water in a pot and heat gently. Stand the jug with the wax and oil in the water. Never put wax straight on the flame.
   Water cannot get much hotter than 100 °C, so the wax stays far below the temperature at which it could catch fire.
6. ADULT: stir gently until everything is melted and clear.

7. ADULT: turn off the heat and lift the jug out with oven gloves. Children may drop in a little candle dye or crayon shavings from a spoon while the adult stirs.
8. ADULT: wait a minute, then add a few drops of fragrance oil and stir.
   Fragrance evaporates from very hot wax; adding it just off the heat keeps more scent in the candle.
9. ADULT: pour the wax slowly into the moulds, keeping each wick in the centre.
10. Leave the candles for 2–4 hours until hard. Do not move them while they set.
    ⚠ **The wax stays hot inside long after the top looks solid. Do not touch or poke it until it is hard and cool.**
11. Trim the wick to about 5 mm. Label your candle with the date and "made from used oil".
    ⚠ **Test-burn one candle with an adult, on a heat-proof plate in a ventilated room. Note any smoke or smell.**

Safety First

⛔ Hot wax and hot oil cause serious burns. Only an adult heats and pours. Children stay an arm's length from the stove.

⛔ Double boiler only. Never heat wax directly over a flame, and never leave melting wax unattended.

⛔ If wax catches fire: turn off the heat and cover it with a pot lid. Never pour water on burning wax or oil.

⚠️ For a burn: cool it under cool running water for 20 minutes and tell an adult. Do not peel off wax stuck to the skin, and do not put butter, toothpaste or ice on a burn.

⚠️ Keep water away from hot wax and oil: even a few drops make hot fat spit and splash.

⚠️ Burn finished candles on a heat-proof plate, away from curtains. Never leave a lit candle alone.

The science

Teens: Paraffin wax is a mixture of long-chain alkanes, solid at room temperature. Cooking oil is mostly triglycerides (glycerol joined to three fatty acids); the C=C "kinks" in unsaturated chains stop the molecules packing tightly, so it is liquid. A 1:2 oil-to-wax blend is softer and melts at a lower temperature than pure wax. In the flame, molten fuel rises up the wick by capillary action, vaporises and burns: hydrocarbon + O_2 → CO_2 + H_2O. With too little oxygen, combustion is incomplete and you get soot (carbon). Filtering matters because solid food particles cannot vaporise, so they char on the wick.

Adults: Fats, oils and grease (FOG) poured into drains solidify in sewers and cause blockages and overflows. Used cooking oil (UCO) has higher-value routes, especially collection for biodiesel, so a candle is best seen as a small upcycling and awareness product rather than the main solution. Be honest with buyers: paraffin is petroleum-derived (plant waxes can replace it at higher cost), and an oil-rich candle may smoke or smell slightly of cooking. Quality control (filtering, ratio, wick size) decides whether it is sellable.

Teacher tip

Filter the oil the day before: paper filtering is slow. One double-boiler station per adult. Pupils weigh, set wicks, choose colours and make labels; adults heat and pour.

Challenge

1. Make three mini-batches with oil : wax = 1:1, 1:2 and 1:3. Compare hardness, smoke and burn time. Which would you sell?
2. Burn rate: weigh a candle, burn it for 1 hour with an adult, weigh again. Burn rate = mass lost ÷ time (g/h). How many hours will a full candle last?
3. Costing: find real prices for wax, wicks and fragrance. Cost per candle = total cost ÷ number of candles. For a 40% margin, price = cost ÷ (1 − 0.40).

Think about it

- Where does used cooking oil go in your home now?
- Why should we never pour oil down the sink?
- What would you change to make your candle better?
- Is a candle the best use for used oil? What else could it become?

🎬 **Video:** wastequest.github.io/#/lab/candle

🐟 Fish-Waste Pet Treats

Fish heads and scraps become crunchy baked treats for cats.

Ages 10+ | ⏱ About 2½ h (adult does the cooking) | Difficulty ★★★ | Adult does hot or sharp steps | 🔥 Needs heat | SDG 12 | SDG 14

> 🧒 **Age and supervision:** An adult handles raw fish, the steamer and the oven. These are occasional treats, not a tested complete pet food.

Waste used: Fish trimmings: heads, skin, flesh scraps · **Product:** Baked cat treats · **Cost:** About RM 1.50 a batch, or RM 4.80 with taurine (2025 student prices).

Why it matters

🌍 **Environment:** Fish heads and scraps are thrown away at markets and in kitchens, where they rot and smell. Food is the biggest part of Malaysian household waste (30.6%); using the scraps cuts it.

💰 **Economy:** Free market leftovers become a product pet owners pay for. The cost table shows where the money goes: the optional taurine powder is most of the cost.

🤝 **Society:** Community cat feeders and animal shelters need cheap, safe treats; this links food-waste reduction to animal welfare.



Ingredients and steps 1–4 (BM). Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- 283 g fresh fish trimmings (heads, skin, flesh scraps), used the same day and kept cold
- About 2 cups (283 g) wheat flour, plus a little for dusting
- 1 egg
- 1 tbsp (14 g) fresh cooking oil, not used oil
- Taurine powder: optional, only if a vet advises
- Steamer, oven, blender, kitchen scale, baking tray and baking paper
- Rolling pin, small cutters or a knife, bowls and spoons
- Gloves, a chopping board used only for raw fish, an airtight container

Ingredient	Amount	Cost (RM)
Fish waste	283 g	Free
Wheat flour	283 g	0.57
Egg	1 (50 g)	0.70
Fresh cooking oil	1 tbsp (14 g)	0.23
Taurine (optional)	11 g	3.30
Total		4.80 (1.50 without taurine)

Prices from the 2025 student poster. The poster's total of RM 5.03 was an adding mistake: the items add up to RM 4.80.

Steps

1. Wash your hands. Wearing gloves, rinse the fish trimmings well. ADULT removes the guts and gills with a knife (they spoil fastest).
   ⚠ **Raw fish can carry bacteria: keep it cold and away from other food, and wash your hands with soap afterwards.**
2. ADULT: steam the fish for about 1 hour until fully cooked.
   Cooking kills germs and parasites and destroys thiaminase, an enzyme in some raw fish that breaks down vitamin B1.
3. Let it cool until it is comfortable to hold. Pick the flesh off and remove EVERY bone. Check twice with your fingers.

4. Blend the boneless fish into a smooth paste (purée). An adult fits, empties and washes the blender: the blades are sharp.
5. ADULT: spread the purée on a lined tray and dry it in the oven at about 117 °C for 15–20 minutes.
6. Mix the dried fish with the egg, flour and oil (and taurine only if a vet said so). Knead into a dough. No salt, onion, garlic or spices!
   ⚠ Onion and garlic are toxic to cats and dogs, even when cooked. The dough has raw egg in it: do not taste it, and wash your hands after kneading.
7. Dust with flour, roll the dough flat and cut tiny shapes (cat-bite size).
8. ADULT: bake at 150 °C for about 20 minutes until firm and dry right through. Thicker pieces need longer.
9. Cool completely. Store in an airtight, labelled container in the fridge and use within about a week. Throw away anything that smells bad or grows mould.
   The 2025 student group saw no mould after 14 days, but home kitchens vary, so we play safe.
10. Give only a few pieces a day, as a treat.
    Vets advise that treats should be no more than 10% of a pet's daily calories.

Safety First

⛔ Steamer, oven, hot trays and knives: adults only.

⛔ Hygiene: wash hands, boards and knives after touching raw fish. Keep raw fish cold until cooking.

⚠️ This is a treat, not a complete diet, and not for humans. Ask a vet first, especially for kittens or pets with kidney problems or allergies.

⚠️ Remove all bones: small bones can choke a pet or cut its mouth.

⚠️ Check pupils' allergies (fish, egg, wheat) before the class.

The science

Teens: Microbes need available water. Drying and baking lower the treat's water activity, slowing bacteria and mould. Heat denatures proteins, killing pathogens and destroying thiaminase. Cats are obligate carnivores: they cannot make enough taurine themselves, and fish is a natural source of it. In the oven, egg proteins set and flour starch gelatinises to bind the dough, and the Maillard reaction between amino acids and sugars gives the brown colour and smell.

Adults: Fish processing leaves large volumes of by-products (heads, frames, skin, viscera). Valorisation routes include fishmeal, fish oil, fish silage, fertiliser and pet food. A home-scale treat demonstrates the idea, but selling pet food brings legal duties (hygiene, labelling, registration): check with the Department of Veterinary Services (DVS) before any commercial step. Nutritionally, a homemade treat cannot replace a complete and balanced diet.

Teacher tip

If time is short, do steps 1–5 (raw fish, steaming, drying) before class; pupils mix, roll, cut and package. Use a separate board and knife for raw fish.

Challenge

1. Cost per treat: count the treats in one batch. Cost per treat = batch cost ÷ number of treats. Compare with a shop-bought cat treat (RM per 100 g).
2. Moisture loss: weigh the purée before and after drying. % water removed = (wet − dry) ÷ wet × 100.
3. Design a label: ingredients in order of weight, storage advice, "treat only, not for humans", and a use-by date.

Think about it

- Which parts of a fish are usually thrown away at the market?
- Why do we dry and bake the treats instead of serving the fish paste?
- Why must the treats never contain onion, garlic or salt?
- Who in your community could use this idea?

🎬 **Video:** wastequest.github.io/#/lab/petfood

🎁 Trash to Treasure: Soap, Tote & Toy Car

Three quick makes: used oil → soap, old T-shirt → bag, bottle caps → racing car.

Ages 8+ | ⏱ 3 activities × about 30 min | Difficulty ★★☆ | Close supervision | 🔥 Needs heat | SDG 4 | SDG 6 | SDG 12

> 🧒 **Age and supervision:** Soap made with lye (sodium hydroxide) is for adults only. Children use melt-and-pour soap base, with an adult doing the melting.

Waste used: Used cooking oil, old T-shirts, bottle caps, ice-cream sticks · **Product:** Soap, tote bag, toy car · **Cost:** Very low: mostly waste. Example shop price (Oct 2026): melt-and-pour soap base RM 26 per kg, so 200 g is about RM 5.20. Glue, rubber bands and decorations extra. Prices vary.

Why it matters

🌍 **Environment:** Textiles are 2.3% of Malaysian household waste, and oil in drains pollutes rivers. Reusing a T-shirt, oil or bottle caps keeps them out of landfill.

💰 **Economy:** Upcycled soap and bags sell at school bazaars; toys from waste cost almost nothing.

🤝 **Society:** Making things builds creativity, problem-solving and pride. Old T-shirts carry memories: "every bag tells a story".



Activity 1: used oil to soap. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

Activity A: soap

- Children: 200 g glycerin melt-and-pour soap base (no lye)
- Optional: 1 tsp dried used coffee grounds (scrub) and 1–2 drops essential oil
- Moulds from waste: yogurt cups, cut milk cartons
- Teacher demo only: 500 g filtered used palm cooking oil, 65 g sodium hydroxide (NaOH), 160 g cold distilled water, goggles, chemical gloves, long sleeves, stainless-steel or heat-proof PP plastic jugs, digital scale, stick blender, lined mould

Activity B: T-shirt tote

- 1 old T-shirt (thick cotton works best)
- Sharp fabric scissors, a ruler and chalk or a marker
- Optional: fabric paint or fabric markers

Activity C: toy car

- 4 bottle caps of the same size
- 2–3 ice-cream sticks and 1 drinking straw
- 2 wooden skewers or long toothpicks (axles)
- 1 rubber band and some glue

Steps

Activity A: soap

1. CHILDREN: an adult cuts the soap base into cubes and melts it in short bursts in a microwave, or in a double boiler.
2. Stir in a pinch of dried coffee grounds and 1–2 drops of essential oil.
   ⚠ **Melted soap base is hot (about 60–70 °C): the adult holds the jug steady while you stir with a long spoon.**
3. ADULT pours it into the moulds. After about 1 hour, pop the soap out. Ready to use!

4. TEACHER DEMO (adults only, full PPE): weigh 160 g cold distilled water into a heat-proof PP or stainless jug. Outdoors or by an open window, slowly add 65 g NaOH TO the water (never water to lye), stirring. It gets very hot and gives off fumes: stand back.
   Lye amount: 500 g × SAP value 0.138 (palm olein) ≈ 69 g; minus about 5% "superfat" so a little oil stays unreacted ≈ 65 g. Always re-check your actual oil in a lye calculator.
5. TEACHER DEMO: when the lye solution and the filtered oil have both cooled to warm (about 40 °C), pour the lye into the oil and stick-blend until it thickens like custard ("trace"). Pour into the lined mould.
6. TEACHER DEMO: cover and leave 24–48 h, unmould with gloves, then cure in an airy place for 4–6 weeks. Fresh lye soap is caustic: do not touch it or take it home until cured. Use UCO soap for laundry and cleaning, not skin.
   Test the cured soap in the Litmus lab: bar soap is naturally alkaline, about pH 9–10.

Activity B: T-shirt tote

7. Lay the T-shirt flat. Cut off both sleeves just inside the seams.
8. Cut a big scoop around the neck to make the bag opening. The shoulder parts become the handles.
9. Turn the shirt inside out. Draw a line about 8 cm up from the bottom hem.
10. Cut strips from the bottom up to the line, about 2–3 cm wide, through both layers.
11. Tie each front strip to the back strip under it with a double knot.
    Then tie one strand of each pair to its neighbour's strand to close the small gaps.
12. Turn the bag right side out: the knots are hidden inside. Decorate it!

Activity C: toy car

13. Cut the straw into two pieces, a little wider than an ice-cream stick. Glue them across the stick, one near each end. These hold the axles.
14. ADULT: make a small hole in the centre of each bottle cap with a bradawl or small drill.
15. Snip the sharp points off the skewers first (ADULT, with strong scissors). Push a skewer through one straw and fix a cap on each end with a dab of glue. Repeat at the other end.
16. Loop the rubber band over the front of the car and hook the other end around the back axle.
17. Roll the car backwards to wind the band round the axle, put it down and let go. Vroom!
    Test, then improve: does it go straight? Do the wheels slip? A thin rubber-band "tyre" on the back wheels adds grip.

Safety First

⛔ Lye (sodium hydroxide, NaOH) is corrosive: it burns skin and can blind. Teacher demo only, with goggles, chemical gloves, long sleeves and good ventilation; children watch from at least 2 metres away. Add lye to water, never water to lye. Never use aluminium containers (they react and make hydrogen gas).

⛔ NaOH looks like sugar or salt. Keep it in its original, labelled container, locked away from children, and never in a food or drink container.

⛔ If lye touches skin or eyes: rinse with plenty of running water for at least 15 minutes (water only: no vinegar). For eyes, get medical help immediately.

⛔ Melted soap base is hot: an adult melts and pours it.

⚠️ Sharp tools: an adult makes the bottle-cap holes. Cut away from your body.

⚠️ Bottle caps are a choking hazard: keep them away from children under 3.

The science

Teens: Saponification: triglyceride + 3 NaOH → glycerol + 3 sodium carboxylates (soap). A soap ion has a polar, hydrophilic head ($–COO^- Na^+$) and a non-polar hydrocarbon tail; in water the tails cluster around grease to form micelles that rinse away. The lye amount uses the oil's SAP value (g NaOH per g oil), minus a small superfat so no free NaOH remains. Toy car: elastic potential energy in the band → rotational kinetic energy of the axle → kinetic energy of the car. Friction at the axles wastes energy; friction at the tyres gives grip.

Adults: The trio shows three circular strategies: chemical conversion (UCO → soap), reuse and repurposing (T-shirt → bag, with no reprocessing energy) and creative reuse of low-value plastics (caps). Cold-process soap is a

chemical process for trained adults; melt-and-pour keeps the session child-safe. Commercially, UCO soap is usually positioned as laundry or cleaning soap; soap sold for use on skin is treated as a cosmetic in Malaysia and needs notification with the NPRA, so check the rules before selling.

Teacher tip

Run the three activities as stations. Do the lye demo only if you are trained; otherwise do melt-and-pour. Collect T-shirts and bottle caps a week ahead.

Challenge

1. Lye maths: with SAP = 0.138 g NaOH per g oil and 5% superfat, how much NaOH does 300 g of oil need? (300 × 0.138 × 0.95 ≈ 39 g)
2. Car race: measure the distance travelled for 1, 2 and 3 winds of the axle. Plot distance against winds. Is it a straight line?
3. Market test: price a T-shirt tote for a school bazaar. What is your cost (materials + time)? What would buyers pay?

Think about it

- Which activity saved the most waste? How do you know?
- Why is lye soap made only by trained adults?
- What other old clothes or toys could you upcycle?

🎬 **Video:** wastequest.github.io/#/lab/treasure

🌈 Say No to Litmus: Rainbow pH From Kitchen Waste

Make your own pH indicator from cabbage leaves or bunga telang: no bought litmus paper needed.

Ages 7+ | ⏱ 45 min | Difficulty ★☆☆ | Close supervision | SDG 4 | SDG 6 | SDG 14

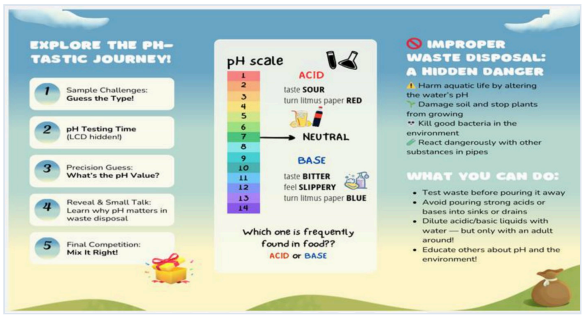
Waste used: Purple-cabbage outer leaves, wilted bunga telang flowers, scrap paper · **Product:** Natural pH indicator and test strips · **Cost:** Almost free: household items.

Why it matters

🌍 **Environment:** Strong acids or bases poured down drains change the pH of rivers and harm fish and useful bacteria. Knowing pH helps us dispose of liquids safely.

💰 **Economy:** A free indicator from kitchen waste replaces bought litmus paper, saving school lab money.

🤝 **Society:** Testing household liquids builds science confidence and helps families think about what they pour away.



pH journey poster. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- 3–4 purple-cabbage outer leaves, OR about 10 bunga telang (butterfly-pea) flowers
- 1 cup (250 mL) water
- A blender, or a bowl and spoon to squash; a sieve or clean cloth
- 6–8 clear cups (or bottles cut in half), labelled
- Samples: lime juice, vinegar, soda water, tap water, salt water, baking-soda water (1 tsp in ½ cup water), soapy water
- Spoons or droppers (washed medicine syringes work)
- Scrap paper or coffee filters for test strips
- Teacher: a pH meter or universal indicator paper to check
- Safety glasses, if available

Steps

1. Tear the cabbage leaves (or flowers) into small bits. Squash them with 1 cup of water for 2 minutes, or an adult runs the blender.
   Warm water, added by an adult, pulls the colour out faster.
2. Strain the liquid through a sieve or cloth. You now have a purple (cabbage) or blue (bunga telang) indicator!
3. Pour a little of each sample into its own labelled cup.
4. Guess first! Write "acid", "neutral" or "base" for each cup.
5. Add 2 spoons of indicator to each cup. Watch the colour change and write it down.
6. Line the cups up from red or pink (acid) through purple or blue (neutral) to green or yellow (base). You made a pH rainbow!
   Roughly, red cabbage goes red (pH 1–2) → pink (3–4) → purple (5–7) → blue (8) → blue-green (9–10) → green (11–12) → yellow (13+). Bunga telang is pink-purple in acid, blue near neutral and green in base.
7. Precision guess (the poster's game): the teacher hides the pH meter's screen. Guess each sample's pH number, then reveal. Who was closest?
8. Mix it right! Put 1 spoon of vinegar and 2 spoons of indicator in a cup. Add baking-soda water a little at a time (a spoon or 1 mL from a syringe), stirring, until the colour turns back to purple or blue (neutral). Count how much you added. Watch it fizz!
   The fizz is carbon dioxide: $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$.
9. Bonus: soak strips of scrap paper or coffee filter in the indicator and let them dry. Now you have your own "litmus" paper!
10. Clean up: these household samples can go down the sink with plenty of water. Leftover cabbage goes to the compost.

Safety First

⚠️ Never taste or touch the test liquids: we look at the colour instead. (The old poster said acids "taste sour" and bases "feel slippery"; we never test that way.)

⚠️ Lime juice and vinegar sting the eyes. Wear glasses if you have them, and rinse with clean water if splashed.

⚠️ Household samples only: no bleach, drain cleaner or toilet cleaner. Never mix cleaning products.

The science

Teens: pH = $-\log_{10}[H^+]$: each unit is a 10-fold change in hydrogen-ion concentration, so pH 3 is 100 times more acidic than pH 5. Anthocyanin molecules gain or lose H^+ as the pH changes, and each form absorbs different wavelengths of light, so we see different colours. Neutralisation: acid + base → salt + water (plus CO_2 when the base is a carbonate or hydrogencarbonate). Counting drops to the colour change is a simple titration.

Adults: pH is a core parameter in waste management: landfill leachate, food-waste fermentation (eco-enzyme is about pH 3.5–4) and industrial effluent are all monitored for pH, and Malaysian effluent regulations set allowed pH ranges for discharges. A zero-cost indicator made from kitchen waste lets community groups screen liquids and teach the concept before investing in meters.

Teacher tip

Make one big jug of indicator before class. Use the hidden-meter game with older pupils; for under-9s, stop at the colour rainbow. Link it to other labs: test the eco-enzyme and the cured soap.

Challenge

1. Colour chart: measure samples with a pH meter and photograph the indicator colour at each pH value.
2. Cabbage vs bunga telang: which shows more colour steps? Which keeps its colour longer?
3. Titration: how many mL of baking-soda solution (from a syringe) neutralise 5 mL of vinegar? Repeat 3 times and average. Predict, then test, what happens with 10 mL. (Expect a lot: household vinegar is much stronger than the baking-soda water.)

Think about it

- Which sample surprised you most?
- Are most foods acidic or basic? Look at your results!
- Why is it dangerous to pour strong acids or bases down the drain?
- How did a "waste" cabbage leaf replace something we usually buy?

🎬 **Video:** wastequest.github.io/#/lab/litmus

☕ Kopi Wira Bau: Coffee-Ground Odour Absorbers

Used coffee grounds become cute shapes that freshen shoes, cupboards and fridges.

Ages 7+ | ⏱ 40 min + 1–2 days drying | Difficulty ★☆☆ | Light supervision | SDG 11 | SDG 12

Waste used: Used coffee grounds from cafés · **Product:** Odour-absorbing shapes · **Cost:** Very low: the cornstarch and corn syrup for one batch cost about RM 0.90–1.00 (shop prices, Oct 2026). Add a little for cinnamon and fragrance. Prices vary.

Why it matters

🌍 **Environment:** Cafés throw away coffee grounds every day; in landfill they rot. Reusing them keeps food waste out of the bin.

💰 **Economy:** A free waste becomes a gift or bazaar product that is cheaper than many shop-bought air fresheners.

🤝 **Society:** Pupils ask a local café for its grounds: a small partnership between school and business.



Materials. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- 1 cup used coffee grounds, completely dried
- ½ cup cornstarch
- 2 tbsp corn syrup (the binder)
- 1 tsp ground cinnamon
- 10–15 drops of fragrance or essential oil (an adult adds them)
- A little water, only if needed
- Tray, bowl and spoon
- Silicone moulds (chocolate or ice-cube shapes)
- A small open container or cloth pouch (an old sock works!)

Steps

1. Ask a nearby coffee shop for its used coffee grounds. Bring a clean container.
2. Spread the grounds thinly on a tray. Dry them for 1–2 days in the sun or an airy place, stirring sometimes. They must be completely dry.
   ⚠ Damp grounds go mouldy within a few days. If you see fuzzy mould, do not sniff it: throw that batch in the compost and wash your hands.
3. In a bowl, mix 1 cup dry grounds, ½ cup cornstarch and 1 tsp cinnamon.
4. Add 2 tbsp corn syrup and the drops of fragrance (adult). Mix well with a spoon.
5. Squeeze a little in your hand. If it crumbles, add water one teaspoon at a time until it holds together.
6. Press the mixture firmly into the silicone moulds.
7. Dry for 24–48 hours until hard, then pop them out.
8. Put them in a small open container or cloth pouch: shoe rack, wardrobe, fridge or bathroom.
9. When the smell fades, dry them in the sun to refresh. Old ones can go into the compost.

Safety First

⚠️ Keep away from pets and toddlers: coffee (caffeine) and essential oils are harmful to dogs and cats. Label them "Not food": they look like chocolate!

⚠️ Essential oils can irritate skin: an adult adds the drops. Wash hands afterwards.

⚠️ Throw away any shape that grows mould. The sugary binder can attract ants: keep shapes in a closed container with small holes, not loose on the floor.

⚠️ In the fridge, use an unscented shape (no fragrance oil) so the food does not pick up the smell.

The science

Teens: Odour molecules are small, volatile compounds. Porous solids trap them by adsorption: molecules stick to a surface (unlike absorption, where something soaks into the bulk). Dried spent grounds have a large internal surface, but far less than activated carbon, which is made at high temperature to open a huge network of pores. So coffee shapes reduce and mask odours rather than neutralise all of them. Cornstarch and corn syrup are the glue: as they dry, sugar and starch molecules form hydrogen bonds and lock the grounds into shape.

Adults: Spent coffee grounds are a steady, clean, single-source waste stream from cafés, ideal for small W2W products. Other proven uses are compost, mushroom-growing substrate and fuel pellets. Keep product claims precise: "helps reduce and mask odours" is honest; "neutralises odours" or "antibacterial" would need testing.

Teacher tip

Collect and dry the grounds a week before class: drying decides success. One batch is enough for a small group; scale up by cups.

Challenge

1. Fair test: put half a cut onion in two closed boxes, one with a coffee shape and one without (try baking soda too). After a day, blind testers rate the smell from 1 to 5.
2. Drying: weigh the wet grounds and the dry grounds. What percentage was water?
3. Price it: cost per shape, then a selling price for a pack of 3 including packaging.

Think about it

- Where else could you use your Kopi Wira Bau?
- Why must the grounds be completely dry?
- Which other café wastes could become a product?

🎬 **Video:** wastequest.github.io/#/lab/odour

🌱 Self-Watering Bottle Garden

A plastic bottle and a T-shirt strip water your plant for you. Teens can add an Arduino brain.

Ages 7+ | ⏱ 40 min (Arduino version: +2 h) | Difficulty ★★☆ | Close supervision | SDG 4 | SDG 9 | SDG 13 | SDG 15

> 🧒 **Age and supervision:** Arduino version: low-voltage parts only (USB or batteries), never mains electricity. An adult checks the wiring before power goes on.

Waste used: PET bottles, old T-shirt · **Product:** Self-watering planter (automatic watering for teens) · **Cost:** Wick version: almost free. Arduino version: example parts (Oct 2026: compatible board, moisture sensor, relay, small pump) add up to about RM 32–57, before wires, tubing and a power supply. Prices vary by shop.

Why it matters

🌍 **Environment:** Bottles are reused before they are recycled, and wick watering gives the plant only the water it needs, with less run-off.

💰 **Economy:** Free planters; automatic watering saves time and water for urban growers.

🤝 **Society:** Small gardens fit flats and school corridors, adding green space where there is little (an idea from a 2024 student project).



How the automatic system works, benefits. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

Wick planter (everyone)

- 1 clean 1.5 L PET bottle, label removed
- 1 strip of old cotton T-shirt, about 2 cm × 25 cm (the wick)
- Potting soil mixed with some compost
- Seeds or seedlings: kangkung, sawi, bayam or chilli
- Scissors or craft knife and a nail or bradawl (adult)
- Paint or markers to decorate

Automatic version (teens)

- Arduino Uno or Nano + USB cable or 5 V power bank
- Capacitive soil-moisture sensor (not a DHT11: that one measures air)
- 5 V relay module
- Small 3–5 V submersible pump with its own battery pack + 1 m silicone tube
- Jumper wires and a covered water container

Steps

Wick planter

1. ADULT: cut the bottle in half, a little below the middle.
2. ADULT: make a hole in the bottle cap with a nail or bradawl.
3. Push the T-shirt strip through the cap hole and tie a knot so it can't slip out. About 10 cm hangs below the cap.
4. Wet the wick. Screw the cap on and turn the top half upside down (cap pointing down) into the bottom half.
5. Fill the top part with moist soil, spreading the wick inside the soil. Plant your seeds.
6. Lift the top out, pour water into the bottom part, and put the top back. The wick must touch the water.
7. Decorate your planter and put it where it gets light. Check the water level every few days.
8. Every week: empty the reservoir, rinse it and refill. This stops Aedes mosquitoes breeding!

Automatic version (teens)

9. Wire it: sensor VCC → 5V, GND → GND, AOUT → A0. Relay IN → D7, VCC → 5V, GND → GND. The pump's battery pack is switched through the relay's COM and NO terminals.

10. Calibrate: open the Serial Monitor and note the reading in dry air and with the sensor tip in a glass of water. Set THRESHOLD between the two.
    Most capacitive modules read higher when dry and lower when wet.
11. Upload the sketch below. Push the sensor into the soil (only up to its line) and test: dry soil → pump runs 2 seconds → waits → reads again.
12. Keep all electronics in a box above the water and log how often it waters.

Arduino sketch (Teens)

```
// WasteQuest self-watering: capacitive sensor on A0, relay on D7
const int SENSOR = A0, RELAY = 7;
int THRESHOLD = 450;          // set from YOUR dry/wet readings

void setup() {
  pinMode(RELAY, OUTPUT);
  digitalWrite(RELAY, LOW);   // pump off (active-LOW relay? swap HIGH/LOW)
  Serial.begin(9600);
}

void loop() {
  int v = analogRead(SENSOR);
  Serial.println(v);
  if (v > THRESHOLD) {        // higher reading = drier soil
    digitalWrite(RELAY, HIGH);  // pump on
    delay(2000);                // 2 s of water
    digitalWrite(RELAY, LOW);   // pump off
    delay(60000);               // let it soak in for 1 min
  }
  delay(1000);
}
```

Safety First

⚠️ Cut bottle edges can be sharp: an adult cuts, and you can cover the edge with tape.

⛔ Electricity and water (Arduino version): use only low voltage (USB, power bank or AA batteries). Never use mains (230 V) power. Keep the electronics above and away from the water, touch wires only with dry hands, and never let the pump run dry.

⚠️ Standing water breeds Aedes mosquitoes (dengue). Keep the reservoir covered and change the water at least once a week: mosquitoes grow from egg to adult in about 7–10 days.

The science

Teens: Capillary action: water molecules stick to the fibres (adhesion) and to each other (cohesion), so water climbs narrow gaps against gravity; the narrower the gap, the higher it rises (Jurin's law, $h = 2\gamma\cos\theta / \rho g r$). As the soil dries and leaves transpire, water keeps moving from the wet wick to the dry soil. In the Arduino version, a capacitive sensor measures the soil's dielectric permittivity: water ($\varepsilon \approx 80$) raises it far above dry soil, so the reading tracks moisture. The code is a feedback loop: measure → compare with a threshold → act → wait → measure again.

Adults: Sub-irrigation (wicking) cuts evaporation and run-off and suits small urban plots; sensor-driven irrigation adds data and saves labour. Costs scale quickly: one 2024 student proposal budgeted RM 51 per pot for a sensor-and-pump vertical garden, so the wick system is the low-cost entry point and automation suits demonstrations or high-value crops. In the waste hierarchy, reusing a bottle comes before recycling it.

Teacher tip

Cut the bottles before class for under-10s. Fast growers (kangkung, sawi) show results within about 2 weeks. Run the Arduino build as a teen club activity.

Challenge

1. Water use: mark the reservoir level every day. How many mL does the plant use per day? Compare a wick pot with a hand-watered pot.
2. Wick test: try cotton, polyester and a sponge strip. How high does coloured water rise in 10 minutes in each?

3. Arduino: print a reading every 10 minutes and graph soil moisture over 3 days. Add a daily limit on pump time to the code.

Think about it

- How does the water get from the bottle to the roots without you?
- Why must we change the water every week?
- What else could you grow in a bottle garden at school?

🎬 **Video:** wastequest.github.io/#/lab/watering

🍊 Eco-Enzyme

Fruit peels + brown sugar + water and a little patience make a natural cleaner.

Ages 7+ | ⏱ 30 min + 3 months fermenting | Difficulty ★☆☆ | Light supervision | SDG 6 | SDG 12 | SDG 13

Waste used: Fruit and vegetable peels · **Product:** Eco-enzyme liquid (mild cleaner, plant feed) · **Cost:** Very low: about RM 0.60 of brown sugar for a batch made with 1 L of water (shop prices, Oct 2026). How much liquid you get back varies.

Why it matters

🌍 **Environment:** Food is 30.6% of Malaysian household waste. Peels turned into eco-enzyme stay out of landfill, where they would rot and release methane.

💰 **Economy:** A bag of sugar plus free peels makes litres of mild cleaner, and the leftover pulp becomes compost.

🤝 **Society:** Easy for every home and school; eco-enzyme groups are active in many Malaysian communities.



Materials: eco-enzyme, soap and mixer. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- 100 g brown sugar, gula merah or molasses (1 part)
- 300 g fresh fruit and vegetable peels, cut small (3 parts): pineapple, orange, papaya, banana…
- 1 litre (1,000 g) clean water (10 parts)
- A 2 L plastic bottle or container with a screw lid (not glass). The mix fills about 1.4 L, so a 1.5 L bottle is too full.
- Kitchen scale, knife, chopping board and funnel
- Label and marker

Steps

1. Wash and chop the peels into small pieces (children use kitchen scissors; an adult cuts hard peels such as pineapple with a knife). No meat, oily food or mouldy fruit.
2. Weigh 1 part sugar, 3 parts peels and 10 parts water: 100 g sugar, 300 g peels, 1,000 g (1 L) water.
   The ratio is by weight. The 2025 poster scaled it up to 1 kg : 3 kg : 10 kg.
3. Pour the water into the bottle, add the sugar and shake until it dissolves.
4. Add the peels. Leave at least a quarter of the bottle empty for the gas.
5. Close the lid. Label it with today's date and the ready date (3 months later).
6. Keep it in a cool, shady place, out of direct sun.
7. For the first 2 weeks, open the lid a little every day to let the gas out, then close it. After that, once a week. Children: ask an adult to open it, slowly, pointing away from faces.
   The 2025 student group built a pressure-relief valve and an Arduino-driven stirrer to do this automatically.
8. After 3 months, strain it. The brown liquid is eco-enzyme! Put the leftover pulp in compost or in your next batch.
9. Always dilute before use. Commonly used: about 1 part eco-enzyme to 10–20 parts water for wiping floors and tables, and much weaker (about 1 : 500–1,000) for watering plants.

Safety First

⚠️ Gas builds up inside: release it daily at first or the bottle can bulge or burst. Never use glass bottles. If the bottle feels hard, open it very slowly over a sink, pointing away from faces: it can spray.

⚠️ Keep it out of hot places such as a car or a sunny window: heat makes the gas build up faster.

⚠️ Not for drinking. Keep it labelled and away from small children. It is acidic: rinse eyes with water if splashed.

⚠️ A thin white film on top is normal. If you see black mould or it smells rotten, add a little more sugar, close it and check again in a week; if it is still bad, compost it and start again.

The science

Teens: This is fermentation. Yeasts turn sugar into ethanol and carbon dioxide ($C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + 2\ CO_2$): that is the gas you release. When air gets in, acetic acid bacteria oxidise ethanol to acetic acid, and lactic acid bacteria make lactic acid. These organic acids lower the pH: studies of 1:3:10 eco-enzyme report about pH 3.5–4. Like mild vinegar, the acids loosen grime and limescale.

Adults: Eco-enzyme is a low-cost, low-tech way to divert fruit waste and engage communities. Be careful with claims: the name suggests strong enzyme activity, but its cleaning action comes mainly from organic acids. It is not a registered disinfectant, and claims that it purifies rivers or treats illness are not supported by solid evidence. Honest positioning (a mild acidic cleaner and a way to cut food waste) protects your credibility.

Teacher tip

Start a class batch early in the term so it is ready before the end. Set a rota for the daily gas release. Test the pH every month in the Litmus lab.

Challenge

1. Measure the pH every 2 weeks for 3 months and plot it. When does it level off?
2. Fair test: one bottle with pineapple peel, one with banana peel. Compare smell, colour and final pH.
3. Cost per litre: price of the sugar ÷ litres made. Compare with a shop floor cleaner (per litre of diluted cleaner).

Think about it

- What made the gas inside the bottle?
- Why do we leave space at the top?
- Which claims about eco-enzyme are true, and which need more evidence?

🎬 **Video:** wastequest.github.io/#/lab/enzyme

🧱 Eco-Bricks & Eco-Bags

Pack soft plastic into a bottle-brick, and weave snack wrappers into a tough bag.

Ages 7+ | ⏱ 45 min a session (one brick takes several sessions) | Difficulty ★☆☆ | Light supervision | SDG 11 | SDG 12 | SDG 13 | SDG 14 | SDG 15

Waste used: Clean, dry soft plastics; PET bottles; foil-lined snack wrappers · **Product:** Eco-bricks (benches, garden borders) and a woven wrapper tote · **Cost:** Free: all waste, plus staples or tape.

Why it matters

🌍 **Environment:** Plastic is 21.9% of Malaysian household waste, and soft plastics like wrappers are hard to recycle. Packed into eco-bricks, they stay out of drains, rivers and open burning.

💰 **Economy:** Eco-bricks replace some bought materials for garden benches and borders; woven bags are durable, waterproof and unique.

🤝 **Society:** Building a school bench together is a visible shared achievement (one 2024 student team gave bags to homeless people).



Eco-brick stool and pillar made by students. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

Eco-brick

- 1 PET bottle (one size for the whole class, e.g. 600 mL or 1.5 L)
- Soft plastics: wrappers, plastic bags, straws, all clean and completely dry
- A packing stick (wooden spoon handle or bamboo stick)
- Scissors and a kitchen scale

Eco-bag (wrapper weave)

- Lots of foil-lined snack or coffee wrappers, washed and dried
- Ruler, pencil and scissors
- Stapler or tape, and 2 long strips for handles

Bottle	Minimum mass	Density
500 mL	167 g	0.33 g/mL
600 mL	200 g	0.33 g/mL
1.5 L	500 g	0.33 g/mL

Global Ecobrick Alliance minimum for building: 0.33 g/mL.

Steps

Eco-brick

1. Wash and dry the plastics. No food, paper, glass, metal or anything sharp.
2. Cut big pieces smaller. Push one coloured piece into the bottom first: it looks nice when you build.
3. Add a little plastic at a time and press it down hard with the stick. Keep packing!
4. Keep going until you can't push any more in. Squeeze the bottle: it should feel like a brick, not a balloon.
5. Weigh it. Target: at least 0.33 g for every mL of bottle: 200 g for 600 mL, 500 g for 1.5 L.
   Density = mass ÷ volume. 200 g ÷ 600 mL = 0.33 g/mL. Lighter bricks are too squishy to build with.
6. Close the cap and write the mass and date on the bottle.

Eco-bag (wrapper weave)

7. Cut the wrappers into rectangles 16 cm × 6 cm.

8. Fold each rectangle in half the long way, open it, fold both long edges to the middle line, then fold in half again. You get a long thin strip.
9. Fold the strip in half across the middle, then fold both ends in to the middle crease. You get a small V-shaped link.
10. Slide the ends of one link into the pockets of another to make a zig-zag chain. Make many chains.
11. Join chains side by side with staples or tape to make two panels, then join the panels into a bag and add two handles.

    One 2024 student team filled small gaps with thin plastic strips. Put shiny sides out for a sparkly bag.

Safety First

⚠️ Clean and dry only: food leftovers make bricks mouldy and smelly.

⚠️ Scissors: cut away from your body. Staples are sharp.

⚠️ Eco-bricks are not for load-bearing walls. Keep them out of strong sun and away from fire: use them inside a frame, or covered with earth or cement, for benches and garden borders.

The science

Teens: Density = mass ÷ volume. Polyethylene and polypropylene, the main soft plastics, have densities of about 0.9–1.0 g/cm³, so a brick at 0.33 g/mL is still roughly two-thirds air. Packing harder raises the density, stiffness and resistance to denting. Soft plastics are hard to recycle because they are thin, light, often dirty and often multi-layer (plastic + aluminium foil), which recyclers cannot separate.

Adults: Eco-bricks are plastic sequestration: they hold plastic out of the environment, but do not recycle it or reduce production. Treat them as the last step after refuse, reduce and reuse. The Global Ecobrick Alliance sets a minimum density of 0.33 g/mL for building use. The woven-wrapper bag (from a 2024 student proposal) upcycles multilayer packaging that has almost no recycling market.

Teacher tip

Agree on ONE bottle size for the whole school so the bricks fit together. A 1.5 L brick takes several sessions: let pupils take bottles home. Keep a class log of the total plastic packed (kg).

Challenge

1. Class total: add up the mass of all the eco-bricks. How many plastic bags is that? (Weigh 10 bags to find the mass of one.)
2. Design: sketch a bench made of 1.5 L eco-bricks. How many bricks do you need? How much plastic would it lock away?
3. Check a real design: a 2024 student stool used 21 bottles of 1.5 L and 800 g of plastic in total. What was the density of each brick? Does it meet the 0.33 g/mL minimum? How much plastic would 21 proper bricks hold?
4. Cost it like an engineer: A 2024 student team spent RM 10.50 on glue sticks and tape for each stool, plus a measuring tape (RM 12) and a glue gun (RM 8) bought once. They planned to sell at RM 25. What is the profit on the first stool? On the tenth?
5. Bag test: load your woven bag with books until it fails. Record the mass. Where did it break, and how could the design be stronger?

Think about it

- Which plastics in your home are hardest to recycle?
- Is an eco-brick recycling? Why or why not?
- How could your school use less soft plastic in the first place?

🎬 **Video:** wastequest.github.io/#/lab/ecobrick

🪱 Composting: Bokashi & Worms

Feed food scraps to microbes and worms and get rich soil back.

Ages 7+ | ⏱ 45 min to set up; 2–8 weeks to finish | Difficulty ★★☆ | Light supervision | SDG 2 | SDG 12 | SDG 13 | SDG 15

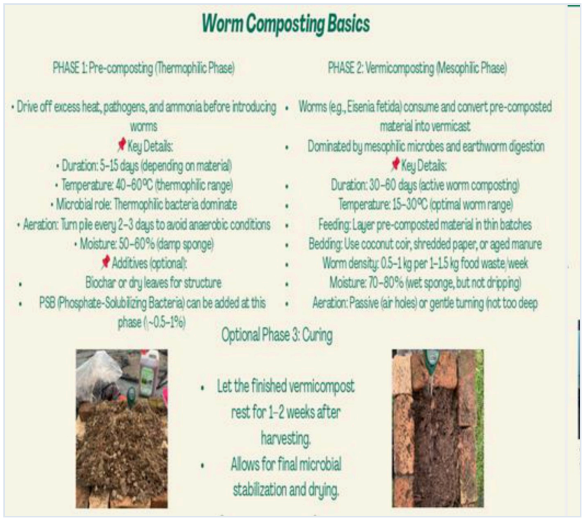
Waste used: Food scraps, dry leaves, cardboard · **Product:** Compost, vermicast and liquid plant feed · **Cost:** Low: reuse old buckets. Example shop prices (Oct 2026): bokashi bran about RM 8–14 per kg. Composting worms are sold by the kilogram and cost much more (one seller: RM 380 per kg in Peninsular Malaysia); the price of a small starter amount varies.

Why it matters

🌍 **Environment:** Food is 30.6% of Malaysian household waste. Buried in landfill without air, it rots into methane (a strong greenhouse gas) and dirty liquid called leachate. Composting turns it back into soil instead.

💰 **Economy:** Compost and worm castings replace some bought fertiliser and potting soil, and gardeners will buy them.

🤝 **Society:** A school compost corner links the canteen, the garden and the classroom; vegetables grown with it can go back to the canteen.



Worm composting poster. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

Bokashi bin

- A bucket with a tight lid, ideally with a tap at the bottom (or two stacked buckets, the inner one with drainage holes drilled by an adult)
- Bokashi bran: about a handful per layer
- Food scraps cut small: fruit and vegetable scraps, cooked rice and noodles, a little cooked meat or fish
- A plate or a bag of sand to press the scraps down
- A garden bed or compost heap for the finished bokashi

Worm bin

- A plastic box with a lid, raised on two bricks over a tray
- Bedding: shredded newspaper or cardboard (not glossy), dry leaves, a handful of garden soil
- Composting worms suited to the tropics, such as the African nightcrawler (Eudrilus eugeniae), from a vermicompost farm. Garden earthworms do not live well in bins.
- Fruit and vegetable scraps, crushed eggshells, used coffee grounds
- A spray bottle of water and gloves

Steps

Bokashi bin

1. Cut the food scraps into thumb-sized pieces and drain off any soup or liquid.
2. Sprinkle a handful of bokashi bran over the bottom of the bucket.
3. Add a layer of scraps, sprinkle bran on top, then press down hard to squeeze out the air.
   Bokashi microbes work without air; air pockets let mould and bad smells grow.
4. Close the lid tightly every time. Repeat the layers until the bucket is full.
5. Every 2–3 days, drain the liquid from the tap. Use it only diluted, at least 1:100 (10 mL in 1 litre of water), to water plants.
6. When full, keep it sealed for 2 weeks. It should smell sour, like pickles. White fluffy mould is fine; black or green mould or a rotten smell means air or too much water got in.
7. Bury it in a trench about 20 cm deep, or mix it into a compost heap. After 2–4 weeks it has turned into soil.
   Bokashi is a fermented "pre-compost": it finishes breaking down in the soil.

Worm bin

8. ADULT: drill small air holes in the lid and sides and drainage holes in the bottom. Stand the box on bricks over a tray.
9. Fill half the box with damp bedding plus a handful of soil. Squeeze it: it should feel like a wrung-out sponge.
10. Gently put the worms on top and leave them a day to settle in. Keep the bin in the shade, never in direct sun or rain.
11. Feed a small amount of chopped scraps at a time, buried under the bedding in a different corner each time. Cover with a handful of dry leaves or paper.
    Aim for about 2–3 parts dry "browns" (leaves, paper) to 1 part wet "greens" (food scraps) by volume.
12. Do not feed worms meat, fish, dairy, oily or salty food, or much citrus, onion, garlic or chilli.
13. After 1–2 months the bottom is dark, crumbly vermicast. Push everything to one side and put fresh bedding and food on the other. In 1–2 weeks the worms move across; scoop out the castings.
14. Mix a little vermicast into potting soil or sprinkle it around your plants.

Safety First

⚠️ Wear gloves or wash your hands well after touching compost, worms or bokashi. Cover any cuts. Open bags of compost, potting mix or bokashi bran away from your face, and do not breathe in the dust.

⚠️ Mould: do not put your face close to a mouldy bin or sniff it. People with asthma or allergies should let someone else open the bin.

⚠️ Bokashi liquid is sour and strong: always dilute it, never drink it, and keep lids shut so flies stay away.

⚠️ Empty the worm-bin tray regularly: standing water breeds Aedes mosquitoes. An adult drills the holes.

⚠️ No pet poo, diseased plants or "compostable" plastic bags: most need hot industrial composting.

The science

Teens: Ordinary composting is aerobic: microbes use oxygen to break down food, releasing CO_2, water and heat. They need carbon for energy ("browns": leaves, paper) and nitrogen to build proteins ("greens": food scraps). About 2–3 browns to 1 green by volume gives roughly the carbon-to-nitrogen ratio of 25–30 : 1 that microbes like; too many greens makes it wet and smelly. Bokashi is anaerobic fermentation: lactic acid bacteria in the bran turn sugars into lactic acid, which makes the bin sour and stops rotting microbes, and the waste finishes breaking down once buried. Worms grind food in their gizzard, and microbes in their gut turn it into castings rich in nutrients that plants can absorb.

Adults: In landfill, food waste decomposes anaerobically and produces methane, a far more potent greenhouse gas than CO_2, plus leachate. Separating food waste at source and treating it at home, school or community level cuts collection costs and landfill load. Bokashi takes cooked food and small amounts of meat (which worm bins and open heaps should not), needs little space and suits flats; worm farms produce high-value vermicast. Note: the 2025 poster's "1:3:2 NPK" was replaced with a browns-to-greens guide, because NPK describes a fertiliser's nutrient content, not a mixing ratio.

Teacher tip

Set up both systems early in the term with a care rota. Buy worms from a local vermicompost farm. Compare after a month: which system handled the canteen waste better?

Challenge

1. Weigh the food scraps your class adds every day for 2 weeks. How many kg would you keep out of landfill in a school year?
2. Fair test: grow two pots of kangkung, one with vermicast mixed in and one without. Measure the height every week.
3. Compare bokashi and worms: cost to set up, space, smell, speed and what each can accept. Which suits your home?

Think about it

- Which foods can go in bokashi but not in a worm bin?

- Why does food buried in landfill make methane?
- How did it feel to look after living worms?

🎬 **Video:** wastequest.github.io/#/lab/compost

🌽 Cornstarch Bioplastic

Cook starch, water, vinegar and glycerin into a plastic you can shape, then test whether it really is "green".

Ages 9+ | ⏱ 60 min + 2 days drying | Difficulty ★★☆ | Adult does hot or sharp steps | 🔥 Needs heat | SDG 9 | SDG 12 | SDG 14

> 🧒 **Age and supervision:** An adult does the heating on the stove. Children measure, stir away from the heat and shape the cooled mixture.

Waste used: Not a waste product: a plant-based swap for single-use plastic · **Product:** Seedling pots, coasters and small shapes · **Cost:** About RM 0.84 per small pot (2024 UPM student estimate; prices may have changed).

Why it matters

🌍 **Environment:** Plastic is 21.9% of Malaysian household waste, and most plastic is made from oil. Starch plastic comes from plants, but "made from plants" does not automatically mean it breaks down anywhere. This lab tests that honestly.

💰 **Economy:** Starch, glycerin and vinegar are cheap. A 2024 UPM student team costed a seedling pot at about RM 0.84.

🤝 **Society:** Students act as materials engineers: they design, test and judge a "green" claim with evidence.



Bio-pots made by students. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- 15 g cornstarch (tepung jagung), about 2 tablespoons
- 100 mL water
- 10 mL (2 teaspoons) white vinegar
- 5–10 mL (1–2 teaspoons) glycerin, from a pharmacy: more glycerin = bendier plastic
- 1–2 drops food colouring (optional)
- Small saucepan, heat-proof spatula, measuring spoons and jug
- Baking paper on a tray, or a small cup or silicone mould to shape a pot
- Stove (adult) and labels

Steps

1. Measure all the ingredients before any heat is turned on.
2. Put the water, cornstarch, vinegar, glycerin and colour into the cold saucepan. Stir until there are no lumps.
3. ADULT: heat on low, stirring all the time.
4. Watch it change: milky white, then thick, clear and gel-like (about 5–15 minutes). Turn off the heat.
   This change is gelatinisation: the starch grains swell, burst and release their long chains.
5. ADULT: spread the hot gel on baking paper about 3–5 mm thick, or press it around the outside of a small cup to make a pot. It is very hot and sticky.
6. When it has cooled to warm (not hot: check with the back of a spoon first), pop any air bubbles with a spoon and smooth the top.
7. Leave it to dry for at least 2 days in an airy place. Turn sheets over after the first day.
8. Peel it off and test it: bend it, stretch it, and put a small piece in water for an hour. What happens?
   Label each sample with its glycerin amount so you can compare.

Safety First

> ⛔ Hot starch gel sticks to skin and burns like hot porridge. Only an adult heats, spreads and moulds it; children mix the cold ingredients and do the testing.

> ⚠️ For a burn: cool it under cool running water for 20 minutes and tell an adult.

⚠️ Not food, even though the ingredients are: label it and do not eat it.

⚠️ Do not put it in the plastic recycling bin: it contaminates real plastic recycling. Small pieces can go in the compost.

The science

Teens: Plastics are polymers: very long-chain molecules. Starch is a natural polymer of glucose made of amylose (mostly straight chains) and amylopectin (branched). Heating starch in water (gelatinisation, at roughly 60–70 °C for corn starch) lets water into the granules, which swell and burst; as the film dries, the released chains tangle and hydrogen-bond into a solid network. Glycerol is a plasticiser: its small molecules sit between the chains and reduce chain-to-chain hydrogen bonding, making the film flexible (too much makes it sticky). Vinegar's role is smaller than often claimed (acid may cut some amylopectin branches for a smoother film), so test with and without it.

Adults: "Bioplastic" mixes two separate ideas. Bio-based means made from plants; biodegradable means microbes break it down under stated conditions. They are independent: bio-based polyethylene is not biodegradable, while some fossil-based polyesters are. ASTM D6866 measures bio-based carbon content (by radiocarbon); it says nothing about biodegradability, which is certified under standards such as EN 13432 or ASTM D6400. Plain starch films are weak and water-sensitive, so they suit short-life uses such as seedling pots, not packaging for wet food. The 2024 student costing of RM 0.84 per pot is a good start; scaling would need consistent drying, moisture resistance and a real end-of-life route.

Teacher tip

Pre-measure the dry ingredients into cups. One saucepan per adult. A microwave also works: heat in 20-second bursts, stirring between (adult only).

Challenge

1. Glycerin test: make films with 5, 10 and 15 mL glycerin. How far does each bend before it cracks? Which would make the best seedling pot?
2. Breakdown test: bury one piece in soil, keep one in a jar of water and one dry. Photograph them every week for 4 weeks.
3. Cost it: price one pot (starch + glycerin + vinegar) and compare with a plastic seedling pot. Would a plant nursery switch?

Think about it

- Is "made from plants" the same as "breaks down in nature"?
- What would you use your bioplastic for, and what should it never be used for?
- How could you make it stronger or more water-resistant?

🎬 **Video:** wastequest.github.io/#/lab/bioplastic

Bottle Vertical Garden

Hang a garden of plastic bottles on a wall or fence: water the top one and it drips down to the rest.

Ages 7+ | ⏱ 60 min to build, then water a little every day | Difficulty ★☆☆ | Close supervision | SDG 2 | SDG 11 | SDG 12 | SDG 13 | SDG 15

Waste used: 1.5 L PET bottles; compost from food waste · **Product:** A hanging vertical garden and mini bottle pots · **Cost:** Almost free: used bottles and string. If you buy soil and seeds, example shop prices (Oct 2026): a 7 L bag of potting mix RM 7.50, a vegetable seed packet RM 5.90. Prices vary.

Why it matters

🌍 **Environment:** Each bottle gets a second life before recycling, and plants on walls cool hot surfaces and give insects food.

💰 **Economy:** Grow kangkung, herbs or cuttings to use or sell, in a space too small for a normal garden.

🤝 **Society:** A green wall makes a school corner or flat balcony nicer, and mini pots make easy gifts.

You need

- 4–6 clear 1.5 L bottles with caps, washed, labels off
- Strong string or nylon cord, and a frame, fence or grille to hang from
- Soil mixed with compost (see the Compost lab)
- Plants: leafy seedlings (e.g. kangkung), herb cuttings, or money plant cuttings
- Craft knife or strong scissors (adult only), a skewer, tape and a marker

Steps

1. Lay a bottle on its side. Draw a window about 15 cm × 6 cm on the top side.
2. ADULT: cut out the window and tape the cut edges so they are not sharp.
3. ADULT: poke 3–4 small drainage holes on the bottom side with a skewer.
   Without holes, water collects at the bottom and the roots rot because they cannot get air.
4. Tie string around both ends (near the neck and the base) to make a hanger. Do the same for every bottle.
5. Hang the bottles one below another, about 25 cm apart, so water dripping from one falls into the next.
6. Fill each bottle two-thirds full with the soil and compost mix.
7. Plant your seedlings or cuttings and press the soil gently around them.
8. Water the top bottle slowly and watch it trickle down. Place the garden where it gets a few hours of sun.
9. Mini pots: cut the bottom 8 cm off spare bottles, poke holes, fill and plant. Tie a ribbon on for a gift.
   The student team also built an Arduino soil sensor and pump to water the garden automatically: see the Smart Watering lab.



Seedlings in mini bottle pots. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

Safety first

⚠️ Cut plastic edges are sharp: an adult cuts, then tape the edges.

⚠️ Wet soil is heavy. Tie the bottles to a strong frame, not above where people walk or sit. An adult hangs anything above head height: never climb on chairs or railings.

⚠️ Water drips down the column: hang the garden away from plugs, sockets and extension cords.

⚠️ No standing water: Aedes mosquitoes can breed in a cap or tray in about a week. Check and empty them weekly.

⚠️ Money plant is for looking at only: it is poisonous to eat and harmful to cats and dogs. Wash hands after gardening.

The science

Teens: Gravity moves water down the column; drainage holes stop waterlogging, because roots need oxygen for respiration. Leaves lose water by transpiration, which also cools the air around them. Growing upwards gives more plants per square metre of floor.

Adults: Green walls reduce surface temperatures and support urban biodiversity, and reusing PET keeps it in use longer before recycling. Sunlight slowly breaks down PET, so expect to replace bottles after some time outdoors, and recycle the old ones.

Teacher tip

Prepare the cut bottles before class. Pupils fill, plant and hang. Make a watering rota and a weekly mosquito check. Pair it with the Compost lab for the soil and the Smart Watering lab for older pupils.

Challenge

1. Measure: how many mL of water do you pour at the top, and how many mL drip out of the bottom bottle? Where did the rest go?
2. Grow the same seedling in a bottle and in the ground. Measure the height every 3 days for 3 weeks and draw a graph.
3. Count how many bottles a class garden of 20 bottles saves from the bin in a year if you replace them once.

Think about it

- Where at home or school could a bottle garden go?
- Is reusing a bottle better than recycling it? Why?
- What would you grow, and who would you give it to?

🎬 **Video:** wastequest.github.io/#/lab/vgarden

🥬 Bottle Hydroponics

Grow leafy vegetables in water, with no soil and no pump, inside a cut plastic bottle.

Ages 9+ | ⏱ 60 min to set up; about 4–6 weeks to harvest | Difficulty ★★☆ | Close supervision | SDG 2 | SDG 6 | SDG 11 | SDG 12

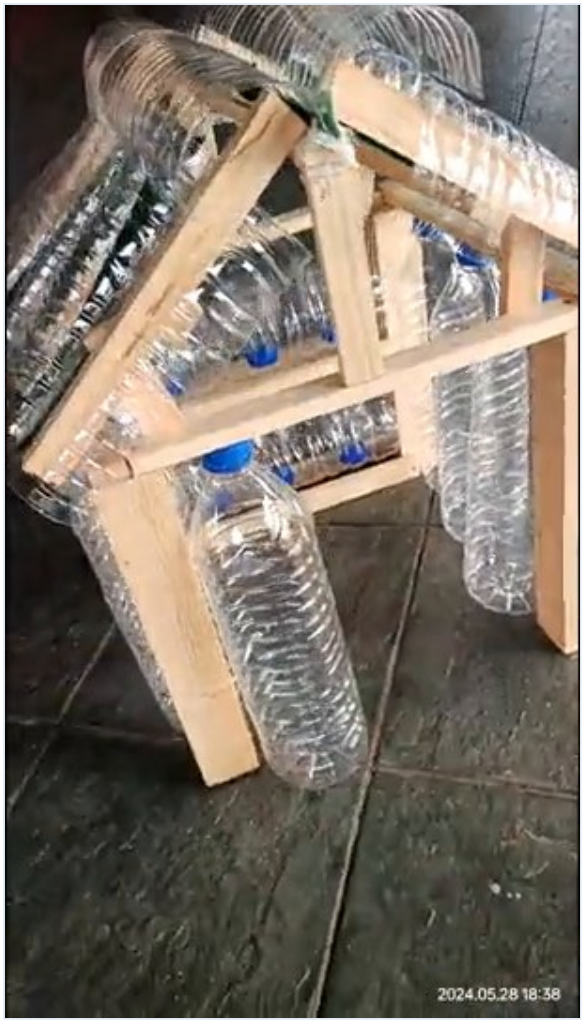
Waste used: 1.5 L PET bottles and caps; scrap wood and bubble wrap for the shelter · **Product:** A soil-free vegetable pot and a mini rain shelter · **Cost:** Bottles are free; hydroponic nutrient (AB mix) and seeds are bought. Example (Oct 2026): a vegetable seed packet RM 5.90; AB mix price varies by pack size (one listing showed RM 16.90).

Why it matters

🌍 **Environment:** Bottles get reused, and a hydroponic pot with no pump uses no electricity. Plants take only the water they need.

💰 **Economy:** Fresh sawi or lettuce at home or in the school canteen, grown on a wall or a corridor with no garden land.

🤝 **Society:** Flats and schools without land can still grow food. A 2024 student team designed a "rain shelter house" so the bottles keep working in the rainy season.



A rain shelter house prototype. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

Bottle pot

- 1 clear 1.5 L bottle per plant, washed
- Seeds of a leafy vegetable (sawi, lettuce or kangkung) and a kitchen sponge cut into 2 cm cubes
- Hydroponic nutrient A and B (AB mix) and a measuring syringe or spoon
- Black paper, foil or old cloth to cover the bottle; tape
- Craft knife (adult only) and a marker

Rain shelter (optional)

- Scrap wood strips, nails and hammer; extra bottles; stapler; clear tape; bubble wrap

Steps

Bottle pot (Kratky method)

1. Sow 1–2 seeds on each wet sponge cube. Keep them damp in a tray until roots and 2 small leaves appear (about a week).
2. ADULT: cut the bottle around one-third from the top. Tape the cut edges.
3. Wrap the bottom part in black paper or foil, leaving a thin window to check the water level.
   Light makes green algae grow in the water. Algae steal nutrients and oxygen from the roots.
4. ADULT: mix the AB nutrient into water exactly as the label says. Add A and B to the water separately, never to each other.
5. Pour the nutrient water into the bottom part. Turn the top part upside down (no cap) and sit it in the bottom like a funnel.
6. Push the sponge with the seedling into the bottle neck so its base just touches the water.
7. Tape around the joint so no light, rain or mosquitoes can get in. Label the bottle "Not for drinking".
8. Keep it in bright light but out of the rain. Do not refill to the top: as the roots grow, the water level drops and leaves an air gap.
   The top roots breathe from the air gap and the lower roots drink. If the water gets very low, top up only to about half.
9. Harvest the leaves in about 4–6 weeks. Pour leftover nutrient water onto garden plants, not into a drain.

Rain shelter house

10. Build a small wooden frame to fit the bottles you collected. Nail each bottle cap to the wood, then screw the bottle into its cap so it stays steady.
11. For the roof, ADULT cuts the top and bottom off bottles and slits them open. Flatten, staple them together and tape the joins so rain cannot leak through.
    The student team used bubble wrap to fill gaps and save cost, and fixed the roof without glue.

Safety First

⚠️ Nutrient salts: an adult measures them. Do not drink or taste, wash hands after, and keep the packets away from small children.

⚠️ Still water can breed Aedes mosquitoes in about a week. Seal every gap with tape and check for wrigglers weekly. If you see any, empty the bottle and start again.

⚠️ Cut bottle edges, nails and hammers: adults cut and hammer, learners wear covered shoes.

The science

Teens: Roots take up nutrients as dissolved ions: nitrogen for leaves, phosphorus for roots, potassium for overall health, plus small amounts of others. Roots also need oxygen. In the Kratky method the water level falls as the plant drinks, so an air gap opens for the upper roots, with no pump needed.

Adults: Non-circulating (Kratky) hydroponics was developed at the University of Hawaii for low-cost growing without electricity. It suits schools and flats, but nutrient solutions are fertiliser: spent solution should go on plants, not into drains, where it feeds algae blooms.

Teacher tip

Sow the sponges a week before the lesson so pupils start with seedlings. One bottle per pair is enough. Keep a class table of the water level and leaf count each week.

Challenge

1. Grow two bottles: one wrapped in black paper, one clear. After two weeks, which has more algae? Which plant is bigger?
2. Mark the water level every 2 days. How many mL does one plant drink a day?
3. Design a rain shelter for 10 bottles on your school wall. Sketch it with measurements and list the waste materials you would use.

Think about it

- Why might hydroponics help people who live in flats?
- What would happen to the roots if the bottle was filled to the top all the time?
- Why did the student team need a roof over their bottles?

🎬 **Video:** wastequest.github.io/#/lab/hydro

☂️ Fused-Plastic Bags & Mini Umbrella

Iron used plastic bags (between baking paper) into a tough new fabric, then make a drawstring bag, a tote or a working mini umbrella.

Ages 9+ | ⏱ 90 min (the umbrella takes a second session) | Difficulty ★★☆ | Adult does hot or sharp steps | 🔥 Needs heat | SDG 12 | SDG 14

> 🧒 **Age and supervision:** Primary pupils (up to 12): an adult does all the ironing as a demonstration, or brings ready-made fused sheets. Children cut, design and sew.

Waste used: Thin plastic carrier bags (marked 2 HDPE or 4 LDPE); plastic straws · **Product:** Fused plastic fabric: drawstring bag, tote bag, mini umbrella · **Cost:** Almost free: used bags and straws, plus thread and string. Baking paper: example rolls from about RM 5.30 (30 cm × 5 m) to RM 9.90 (38 cm × 12 m); prices vary.

Why it matters

🌍 **Environment:** Carrier bags are light, blow into drains and are rarely recycled. Fusing them turns many weak bags into one strong, waterproof sheet.

💰 **Economy:** Waterproof pouches and totes from free waste can be sold at school fairs.

🤝 **Society:** Sewing and designing build real skills, and each bag carries a message about plastic.



Mini umbrella frame made from ironed straws. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- 5–10 used carrier bags, clean and dry (check the symbol: 2 or 4 only)
- Baking (parchment) paper, a clothes iron and an ironing board or towel-covered table
- Scissors, ruler and marker
- Needle and thread or a sewing machine; buttons
- For the umbrella: about 10 plastic straws

Steps

Make the fabric (ADULT irons)

1. Cut off the handles and the bottom seam of each bag, then cut down one side so it opens into a flat sheet.
2. Stack 4–6 layers of bag on baking paper and cover with another sheet of baking paper. Plastic must never touch the iron.
3. ADULT: iron on a low-medium setting with no steam, keeping the iron moving for 15–20 seconds. Turn over and repeat.
   ⚠ Too cool and the layers will not join (as one student team found); too hot and it shrinks, gets holes and smells. Stop if it smokes.
4. Let it cool, then peel off the paper. The bags have shrunk into one stiff, waterproof sheet.

Drawstring bag

5. Measure, mark and trim the fabric to two equal rectangles.
6. Fold 2 cm over at the top edge, cover with baking paper and ADULT irons the fold to seal a tunnel. Then seal the sides and bottom the same way, or sew them.
7. Make a plastic-bag cord: fold a bag 3 times, cut strips two fingers wide, open them out and cut in a zig-zag to get one long strip. Thread it through the tunnel and knot.

Tote bag

8. Cut fabric pieces to the size you want (front, back, base, two handles). Sew them together by hand or machine and add a button.

Mini umbrella

9. ADULT irons straws flat between baking paper to make stiff strips. Cut each strip in half for 2 ribs, or in quarters for 4 stretchers.
   Most straws are polypropylene (code 5), which softens at a higher temperature than bag plastic. Raise the heat a little at a time, keep the window open and stop if it smokes.
10. Bundle 4 straws as the shaft. Make 2 small holes at the ends of each rib and stretcher.
11. Sew the stretchers to a short straw sleeve (the runner) with needle and thread, and the ribs to the top of the shaft. Add a stopper so the runner locks open.
12. Open and close it to test, then cut a circle of fused fabric as the canopy and stitch it to the rib tips.

Safety First

⛔ ADULT ONLY for all ironing. Open windows or use a fan, keep the iron on low-medium, always use baking paper on both sides, and stop at once if there is smoke or a sharp smell.

⛔ Use only bags marked 2 (HDPE) or 4 (LDPE). Never heat PVC (3), foil-lined wrappers or bags with metal print: they can give off toxic fumes.

⚠️ Hot plastic sticks to skin and burns. Let it cool before touching. Stand the hot iron upright, out of reach, and switch it off when you finish.

⚠️ Needles and scissors: count needles before and after, and keep fingers clear of the sewing-machine needle.

The science

Teens: Polyethylene is a thermoplastic: long chain molecules that slide past each other when warm. Under the iron, chains at the surfaces mix across the layers, and on cooling they lock together. LDPE softens and melts at a lower temperature than HDPE, which is why a low-medium iron is enough and a hot iron burns holes.

Adults: Fusing is upcycling by heat, so the controls matter: thin PE melts at roughly 115–135 °C, well below the temperature at which it decomposes, but PVC releases hydrogen chloride when heated and printed or foil layers can release other fumes. Ventilate, keep temperatures low and sort by resin code first. Once fused and sewn, the product can still go back into PE recycling at end of life if no other materials are attached.

Teacher tip

The adult runs one ironing station by an open window; groups queue with their stacked bags and baking paper. Others cut, design and sew. Make the fabric in session 1 and the umbrella in session 2.

Challenge

1. Strength test: hang a bag of 1 layer, 4 layers and 8 layers from a hook and add water bottles until each tears. Plot layers vs. load.
2. Waterproof test: pour 100 mL of water into your drawstring bag. Does any leak out in 5 minutes? Where?
3. Count how many carrier bags went into your product. How many would your whole class save in a term?

Think about it

- Why must plastic never touch the iron directly?
- Is a fused-plastic bag better than refusing the carrier bag in the first place?
- What else could you make from the fabric?

🎬 **Video:** wastequest.github.io/#/lab/fused

🛟 Bubble-Wrap Float Ring (Model)

Roll parcel bubble wrap into a ring that floats, then test how much weight it can hold up in a basin.

Ages 7+ | ⏱ 45 min | Difficulty ★☆☆ | Close supervision | SDG 11 | SDG 13 | SDG 14

> 🧒 **Age and supervision:** This is a model for learning how things float. It is not a safety device: never use it in water to hold up a person.

Waste used: Bubble wrap from parcels; old PE tarpaulin or plastic bags · **Product:** A model float ring for a buoyancy experiment · **Cost:** Free: parcel bubble wrap, a scrap of tarpaulin or a plastic bag, and tape.

Why it matters

🌍 **Environment:** Online shopping produces lots of bubble wrap and plastic film. Kept in use, it stays out of drains, where plastic blocks water flow and makes flash floods worse.

💰 **Economy:** Free materials make a full science experiment on floating and sinking.

🤝 **Society:** Floods are Malaysia's most common natural disaster. Building a model opens a talk about flood safety: what to do, and what never to do.



4 steps: layer, roll, join, cover. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- A long piece of bubble wrap (about 1 m × 30 cm for a model)
- Wide tape (packing or duct tape)
- An old PE tarpaulin scrap or 2 plastic bags for the cover
- A basin or big tub of water, and weights: 500 mL water bottles, coins or marbles
- Kitchen scale and ruler

Students' full-size design	Value
For a child up to	30 kg
Inner radius	0.25 m
Outer radius	0.57 m
Tube radius	0.16 m
Volume	≈ 0.2 m³

From the 2024 lifebuoy team's slides. A model, not a tested safety device.

Steps

1. Lay 3 layers of bubble wrap on top of each other and tape them together into one thick sheet.
2. Roll the sheet tightly into a long tube and tape along its length.
3. Bend the tube into a ring and tape the two ends firmly together.
4. Cover the ring with strips of tarpaulin or plastic bag and tape them down. This is your waterproof skin.
5. Weigh the ring. Measure its outer and inner diameter and the thickness of the tube.
6. Float it in the basin. Add weights one at a time until water reaches the top of the ring. Record the total mass it held.

   1 litre of water has a mass of 1 kg. A ring that can push aside 2 litres of water can hold up about 2 kg, minus its own mass.

Safety First

> ⛔ This is a learning model, NOT a life-saving device. Never use it for swimming, in a flood or for a rescue. Only certified lifejackets and lifebuoys are safe.

> ⛔ In a flood: stay out of the water, move to higher ground and call 999. If someone is in the water, REACH or THROW, don't GO in.

⚠️ An adult stays by the basin the whole time: young children can drown in a few centimetres of water. Empty it after use.

⚠️ Keep bubble wrap and plastic bags away from babies and toddlers (suffocation).

The science

Teens: Archimedes' principle: the upward force equals the weight of water pushed aside. A ring is a torus, with volume $V = 2\pi^2 R r^2$, where R is the distance to the centre of the tube and r is the tube radius. Each litre under water gives about 1 kg of lift. For the students' design ($R = 0.41$ m, $r = 0.16$ m), $V \approx 0.2$ m³ = 200 L.

Adults: Volume is not the only test of a life-saving device. A certified ship's lifebuoy (IMO Life-Saving Appliances Code) must hold up at least 14.5 kg of iron in fresh water for 24 hours, survive a drop into water and resist fire, oil and sunlight. Bubbles can burst and seams can leak, which is why DIY floats stay as models. The real lesson is prevention: plastic kept out of drains reduces flood risk.

Teacher tip

Run it as a fair test: groups change one thing (layers, tube thickness or ring size) and compare the mass held. Finish with the flood safety rules and "Reach or throw, don't go".

Challenge

1. Use $V = 2\pi^2 R r^2$ to predict how much your model should hold. Compare with your test. Why are they different?
2. Check the students' numbers: with $R = 0.41$ m and $r = 0.16$ m, what volume do you get?
3. Make a poster for your neighbourhood: "Keep plastic out of drains" and the flood safety rules.

Think about it

- Why is a home-made float not safe for a real flood?
- How does plastic in drains make floods worse?
- What would you do if you saw someone in floodwater?

🎬 **Video:** wastequest.github.io/#/lab/lifebuoy

🛌 3-in-1 Bubble-Wrap Sleeping Bag

Sew squares of bubble wrap and old cloth into a mat that becomes a blanket or a buttoned sleeping bag.

Ages 10+ | ⏱ About 2 hours, over 2–3 sessions | Difficulty ★★☆ | Close supervision | SDG 1 | SDG 3 | SDG 11 | SDG 12

> 🧒 **Age and supervision:** A learning prototype only: it has not been tested for outdoor or emergency use. An adult supervises needles and any sewing machine.

Waste used: Parcel bubble wrap; old bedsheets, curtains or T-shirts · **Product:** A 3-in-1 mat, blanket and sleeping bag · **Cost:** Almost free: bubble wrap from parcels and old cloth; thread and buttons.

Why it matters

🌍 **Environment:** Bubble wrap from online shopping is light, bulky and rarely recycled. Here it gets a long second life with old cloth that would also be thrown away.

💰 **Economy:** Two free waste streams make a useful product for camping, school trips or flood relief centres.

🤝 **Society:** A 2024 student team designed it for people sleeping rough on cold, hard floors: a gift that shows care.



The rolled-up bubble-wrap sleeping bag. Photos and posters: UPM chemical engineering students, Faculty of Engineering. Lab guides checked and corrected by the WasteQuest team.

You need

- Bubble wrap, enough for 18 squares of 30 cm × 30 cm
- Old cloth (bedsheet, curtain or T-shirts) for 18 matching squares
- Ruler, marker, scissors and pins
- Big needle and strong thread, or a sewing machine
- About 10 buttons

Steps

1. Cut 18 squares of bubble wrap and 18 squares of cloth, each 30 cm × 30 cm.
2. Pin a cloth square onto each bubble-wrap square, bubbles facing in.
3. Sew the pairs into 6 rows of 3 squares.
4. Sew the 6 rows together into one sheet, about 90 cm × 180 cm.
   The student team made theirs about 100 cm × 180 cm: big enough for an adult to lie on.
5. Fold a strip of cloth over the edges and sew it down so no bubble wrap sticks out.
6. Sew buttons along one long side and the bottom. Cut small button holes (or sew loops) on the matching edges.
7. Test all 3 ways: open flat as a mat (bubble side down), wrap as a blanket, or fold in half and button up as a sleeping bag.
   The student team's next idea: velcro instead of buttons, so it is quicker to change.

Safety First

> ⛔ Plastic over the face can stop breathing. Never let babies or toddlers use or play with it, and never cover the face.

> ⚠️ Plastic burns and melts easily: keep it away from candles, mosquito coils and stoves.

> ⚠️ Needles and pins: count them before and after. Keep fingers clear of a sewing-machine needle.

The science

Teens: Heat moves by conduction, convection and radiation. Lying on a floor loses heat mainly by conduction. Still air conducts heat about 40 times less well than concrete, and sealed bubbles stop the air moving (no convection), so the bubble layer is a good insulator. The plastic also blocks damp from the ground.

Adults: This combines two hard-to-recycle streams, LDPE film and mixed textiles, into a long-life product. Plastic does not breathe, so sweat can condense inside: the cloth layer faces the body. Design questions for a real relief

product: washability, fire behaviour and how it is collected at end of life.

Teacher tip

Split the class: cutters, pinners and sewers. Hand-sewing suits ages 10+; a teacher or older pupils use the machine. Two classes can make one bag each for a local shelter or relief centre (ask them first what they need).

Challenge

1. Warmth test: put a cup of warm water on a square of bubble wrap and another on plain cloth on a cold floor. Measure the temperature every 5 minutes. Which cools faster?
2. Area maths: how many 30 cm squares do you need for a 100 cm × 180 cm bag? (Remember the seams.)
3. Research plastic-bag sleeping mats crocheted from strips of hundreds of bags (the student team's inspiration). Compare time and materials with your bag.

Think about it

- Who in your community could use this, and how would you ask them?
- Why is still air a good insulator?
- What other waste could you use instead of bubble wrap?

🎬 **Video:** wastequest.github.io/#/lab/sleepbag

8. Teacher Guide & Certificates

📋 Overview & outcomes

WasteQuest turns waste education into a quest. Learners find out where Malaysia's waste goes, play games that build sorting and science skills, then make real products from waste in hands-on labs. The module combines gamification (points, badges and certificates), experiential learning, STEM and sustainability education. Its aim is to grow **Zero-Waste Champions**: people who reduce, sort and transform waste at home, at school and at work.

📥 **Downloads:** Booklet (English, PDF) · Booklet (BM, PDF) · 📖 eBook (read online) · eBook (English, PDF) · eBook (BM, PDF) · Offline pack with videos (ZIP, about 215 MB): unzip to a laptop or USB and open WasteQuest_offline.html, no internet needed · 🔗 QR code, website embed and Google Classroom links

What is new in the 2026 edition

- A website that runs on phones, tablets and classroom projectors, in English and Bahasa Melayu.
- Four levels: kids (7–12), teens (13–17), adults, and a teacher view with teaching notes.
- Games, science simulations, 15 hands-on labs with safety controls, a live Class Battle and three certificate tracks.
- Plastic-reuse labs: vertical garden, bottle hydroponics, fused-plastic bags, float ring model and bubble-wrap sleeping mat.
- Science checked and corrected: unsafe or inaccurate methods from earlier student proposals are not used.

Teaching approach

🎮

Gamification
Points, badges, levels and certificates keep learners motivated and give instant feedback.

🔁

Experiential learning
Learners do, reflect, understand and try again (Kolb's cycle, below).

🔬

STEM
Chemistry (pH, fermentation), biology (microbes, compost), maths (data, costing) and engineering design.

🌏

Sustainability and ESG
Malaysian waste data, the SDGs, Act 672 and Extended Producer Responsibility (EPR).

Kolb stage	What learners do in WasteQuest
1. Concrete experience	Play a game; make a product in a lab.
2. Reflective observation	Quick checks, class discussion, reflection worksheet.
3. Abstract conceptualisation	Learn chapters and science simulations explain why it works.
4. Active experimentation	Improve the product, write an action plan, pitch a W2W idea.

Objectives

1. Raise awareness of Malaysia's waste challenge and the waste hierarchy through interactive activities.
2. Introduce Waste-to-Wealth and the circular economy through experiential, hands-on learning.
3. Strengthen STEM learning through safe, science-based waste conversion projects.
4. Encourage sustainable lifestyles, environmental stewardship and green entrepreneurship in daily life.

Learning outcomes, mapped to activities and assessment

By the end of the module, learners will be able to... The last column shows which of the four 2025 learning outcomes each one extends.

LO	Outcome	Level	Activities	Assessment	2025 LO
LO1	Explain sustainability, the waste hierarchy, Waste-to-Wealth (W2W) and the circular economy, using Malaysian waste data.	Understand	📘 **Learn 1** 📘 **Learn 2** 📘 **Learn 3** 📘 **Learn 4** 🕵️ **Myth or Fact?** ✏️ **Let's Cross Together!**	Learn quick checks; quiz	LO1
LO2	Separate household waste correctly into the SWCorp blue, orange and brown bins and explain why separation at source matters.	Apply	📘 **Learn 5** 🗑️ **Sort It Out!** 🃏 **Eco-Match** 🔎 **Fun Puzzle: Word Search**	Sort It Out score; quiz	LO4

LO	Outcome	Level	Activities	Assessment	2025 LO
LO3	Explain the science behind a waste conversion: composting, fermentation, natural pH indicators or bioplastics.	Understand, analyse	🌱 **Compost Master** 🍊 **Eco-Enzyme Mixer** 🧪 **Say No to Litmus!** 🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste**	Reflection worksheet; quiz	LO1
LO4	Make a W2W product safely by following a method and its safety controls.	Apply (hands-on)	🔬 **All 15 labs**	Product rubric; lab badge	LO2
LO5	Evaluate the environmental, economic and social value of a W2W product, including its cost and price.	Evaluate, create	💰 **Trash-to-Cash Tycoon** 👣 **My Waste Footprint** 📘 **Learn 4** 📘 **Learn 6** 🎤 **W2W pitch form**	W2W pitch rubric	LO2
LO6	Reflect on personal habits and promote zero-waste practices at home, in school and in the community.	Value (affective)	📘 **Learn 7** 🥁 **Class Battle**	Action plan; reflection rubric	LO3, LO4

Benefits of Waste-to-Wealth

🌿 Environmental	💰 Economic	🤝 Social
Less pollution of air, rivers and sea	Value-added products from low-cost waste	Greater environmental awareness
Lower greenhouse gas emissions, especially methane from landfills	Green entrepreneurship and small businesses	Sustainable lifestyles and habits

🌿 Environmental	💰 Economic	🤝 Social
Conserves raw materials, water and energy	Lower disposal costs; new jobs	Creativity, innovation and teamwork

🗓️ Session plans

Four ready-to-run plans. Each activity links to the page you need; the right-hand column gives a question to ask. Adjust timings to your group.

⏱️ 60-minute lesson

Kids 9–12 or teens · one class period (split into two if periods are 30–40 min)

Materials: Projector; three boxes coloured blue, orange and brown; 10 clean waste items; natural pH indicator prepared by the teacher; small cups of safe test liquids (vinegar, lime juice, baking-soda water, soapy water); exit tickets.

Time	Activity	Open	Facilitator prompt
0:00–0:05	Hook: "What did you throw away today?" Show the 1.17 kg stack.	📘 Learn 1	*"If each of us throws away about 1.17 kg a day, how much does our class throw away in a year?"*
0:05–0:20	Learn chapters 1 and 5 on the projector; answer the quick checks by show of hands.	📘 Learn 1 📘 Learn 5	*"Which bin is for paper? Why must food waste stay out of the recycling bins?"*
0:20–0:35	Play Sort It Out in pairs, or a 10-question Class Battle.	🗑️ Sort It Out! 🏟️ Class Battle	*"Before you drop it, say the bin and the reason."*
0:35–0:52	Hands-on demo: groups test safe liquids with the natural pH indicator and record the colours, then try the pH lab.	🌈 Say No to Litmus: Rainbow pH from Kitchen Waste 🧪 Say No to Litmus!	*"Which liquids are acids? How can a plant scrap tell us?"*
0:52–1:00	Exit ticket: one thing I learned and one action I will take.	📘 Learn 7	*"What will you refuse or reduce this week?"*

⚠️ **Adult supervision required.** The teacher prepares the indicator with hot water before class. Learners use only the safe liquids listed: never bleach, drain cleaner or oven cleaner. Do not taste anything.

🛠️ Workshop: 2 hours 20 minutes

Teens, university students or community groups · the original 2025 module flow, improved

Materials: Projector and internet (for Class Battle); devices for games (one per pair); lab materials from each lab page; printed product rubrics and reflection worksheets; first-aid kit; gloves, aprons, paper towels.

Time	Activity	Open	Facilitator prompt
0:00–0:15	Briefing: welcome and objectives; groups of 4–5 with roles (leader, materials, safety officer, recorder, presenter); safety briefing with the lab rules checklist.	📎 **Safety**	*"Our safety officer can stop the group at any time if something looks unsafe."*
0:15–0:35	Presentation: waste in Malaysia, the waste hierarchy, circular economy and W2W.	📘 **Learn 1** 📘 **Learn 2** 📘 **Learn 3** 📘 **Learn 4**	*"Where does most household waste in Malaysia end up? How could it become a resource?"*
0:35–1:05	Quiz and games: station rotation (8 min each) of Sort It Out, Memory Match and Myth or Fact, then a 6-minute Class Battle.	🗑️ **Sort It Out!** 🃏 **Eco-Match** 🕵️ **Myth or Fact?** 🥁 **Class Battle**	*"Which myth surprised you the most?"*
1:05–2:05	Hands-on: each group completes one lab and scores it with the product rubric. Low-heat choices: ecobricks, natural pH indicator paper, odour absorber. Candle or bioplastic only with an adult at each stove.	🧱 **Eco-Bricks & Eco-Bags** 🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste** ☕ **Kopi Wira Bau: Coffee-Ground Odour Absorbers** 🕯️ **Eco-Candles from Used Cooking Oil** 🌽 **Cornstarch Bioplastic**	*"What waste did you use, what did you make, and what science made it work?"*
2:05–2:20	Reflection and assessment: a 1-minute pitch per group, the reflection worksheet, and the quiz or certificate quest.	⚡ **Eco Quiz Blast** 🏅 **Certificates**	*"Who would use or buy your product, and how much waste could it save?"*

⚠️ **Adult supervision required.** Heat (candle, bioplastic) and sharp tools need an adult at every station. Prepare materials and cut bottles before the session.

🌱 Eco-Club: 4 weeks

Primary or secondary school club · one 60–90 minute meeting a week

Materials: Gloves and a kitchen scale for the bin audit; a plastic bottle for eco-enzyme; a compost bin; lab materials; space for a small exhibition.

Time	Activity	Open	Facilitator prompt
Week 1	Waste detectives: Learn 1–2; Sort It Out; audit one day of the class bin (weigh by type, gloves on); start an eco-enzyme bottle, which needs 3 months.	📘 **Learn 1** 📘 **Learn 2** 🗑️ **Sort It Out!** 🍊 **Eco-Enzyme**	*"What surprised you in our bin? Which item could have been avoided?"*
Week 2	Science of change: Learn 3–4; compost, eco-enzyme and pH simulations; natural pH indicator lab; set up a compost bin.	📘 **Learn 3** 📘 **Learn 4** 🌱 **Compost Master** 🍊 **Eco-Enzyme Mixer** 🧪 **Say No to Litmus!** 🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste** 🪱 **Composting: Bokashi & Worms**	*"What do microbes need to break down food waste?"*
Week 3	Makers: one product lab per group, then cost the product in Waste to Cash.	🎁 **Trash to Treasure: Soap, Tote & Toy Car** 🌱 **Self-Watering Bottle Garden** 🌽 **Cornstarch Bioplastic** 🕯️ **Eco-Candles from Used Cooking Oil** 💰 **Trash-to-Cash Tycoon**	*"What does one item cost to make? Who would buy it, and for how much?"*
Week 4	Champions: Learn 5–7; action plans; Class Battle tournament; mini exhibition for parents or the school; certificate quests.	📘 **Learn 5** 📘 **Learn 6** 📘 **Learn 7** 🥁 **Class Battle** 🏅 **Certificates**	*"What will our club do next term to keep the change going?"*

⚠️ **Adult supervision required.** Eco-enzyme bottles build up gas: use plastic bottles only, never sealed glass, and release the gas daily in the first month. Cover any standing water so Aedes mosquitoes cannot breed.

💼 Adult / ESG half-day: 3.5 hours

Staff, community leaders, CSR and sustainability teams · counts towards the Practitioner Certificate of Completion (non-credit)

Materials: Projector and internet; one device per pair; lab materials for three labs; printed pitch rubric; aprons and gloves.

Time	Activity	Open	Facilitator prompt
0:00–0:20	Opening: Malaysia's waste picture and why it matters to organisations (set the audience to Adults).	📘 **Learn 1**	*"What waste does your organisation produce, and who pays for it?"*
0:20–0:50	Circular economy, the waste hierarchy and Extended Producer Responsibility (EPR).	📘 **Learn 2** 📘 **Learn 3** 📘 **Learn 4**	*"Where on the hierarchy are your current practices? What would moving one step up take?"*
0:50–1:20	Simulations in pairs: carbon footprint and Waste to Cash costing.	👣 **My Waste Footprint** 💰 **Trash-to-Cash Tycoon**	*"What price makes the product viable without overclaiming its green benefits?"*
1:20–1:35	Break		
1:35–2:35	Hands-on lab: choose one of three.	🕯️ **Eco-Candles from Used Cooking Oil** 🍊 **Eco-Enzyme** 🌽 **Cornstarch Bioplastic**	*"What would it take to make 100 of these a month, safely?"*
2:35–3:00	Pitch preparation with the W2W pitch form.	🎤 **W2W pitch form**	*"State the waste stream, the customer, the unit cost and one SDG."*
3:00–3:20	Pitches (3 minutes each) with peer scoring on the pitch rubric. For more than 6 groups, run a gallery walk.	📎 **Certificates**	*"What is the strongest point of this pitch, and one question you would ask an investor?"*
3:20–3:30	Close: Practitioner quest (20 questions) now or as follow-up; next steps.	🏅 **Certificates**	*"What one change will your team make next month?"*

⚠️ **Adult supervision required.** Hot wax and hot starch mixtures cause burns: one facilitator per stove, long sleeves and closed shoes.

🎓 Certificates

WasteQuest offers three certificate tracks. Learners complete them on this website at 🏅 **Certificates**. Every certificate is a **Certificate of Completion** with a verification code.

Track	Audience	Assessment	Pass mark	Other requirement
🌱 **Junior Eco-Hero**	Kids 7–12	10 questions from the kids track	**70%**	Earn at least 3 game badges
🏆 **Zero-Waste Champion**	Teens 13–17	15 questions from the teens track	**75%**	At least 5 badges, including at least 1 lab badge
🎓 **Waste-to-Wealth Practitioner**	Adults	20 questions from the adults track	**80%**	At least 2 lab badges + W2W product pitch (portfolio, teacher-verified)

Non-credit Certificate of Completion. No MQA accreditation, quality verification, academic credit or credit transfer is claimed. A formal UPM micro-credential would be a separate route that needs approval through the Faculty and UPM's Centre for Academic Development and Leadership Excellence (CADe-Lead).

Practitioner design: 20 notional learning hours (non-credit)

In Malaysia, 1 credit = 40 notional learning hours (NLH): all the time a typical learner needs, including guided and independent learning and assessment. Twenty hours of learning does not by itself give any credit; credit needs an approved academic route. The Practitioner is designed with reference to the MQA *Guidelines to Good Practices: Micro-credentials* (2020).

Component	Mode	NLH	Evidence
Online learning: Learn chapters 1–7 (adult view)	Independent	4	All 7 quick checks complete (Learn badge)
Games and simulations (Footprint, Waste to Cash, pH, compost, eco-enzyme)	Independent or guided	3	Game badges
Hands-on labs (at least 2, incl. preparation and clean-up)	Face-to-face, supervised	6	Lab badges; product photos and notes
Portfolio and W2W product pitch (research, costing, rehearsal)	Independent and group	5	Completed pitch form
Assessment: final quest (20 questions) and pitch session	Online and face-to-face	2	Quest score; pitch rubric

Component	Mode	NLH	Evidence
Total		20	non-credit

Course learning outcomes (2) and assessment matrix

Two course learning outcomes (CLOs), each covering module LOs as sub-outcomes. This follows good practice of a small number of focused outcomes.

CLO	Sub-outcomes	Activities	Assessment
CLO1 Explain W2W, circular-economy and waste-management principles in the Malaysian context (Act 672, EPR, SDGs).	LO1, LO2, LO3	📘 Learn 1 📘 Learn 2 📘 Learn 3 📘 Learn 4 📘 Learn 5 📘 Learn 6 👣 My Waste Footprint	Final quest, 20 questions (pass 80%)
CLO2 Produce and pitch a safe, costed W2W product that adds environmental, economic and social value.	LO4, LO5, LO6	🔬 Labs 💰 Trash-to-Cash Tycoon 🎤 W2W pitch form	Lab portfolio (2 labs) and W2W pitch rubric

Assessment weighting and completion rule

Assessment	Weight	Minimum to complete
Final quest (20 questions, adults track)	30%	80%
Lab portfolio (2 labs, product rubric)	30%	2 lab badges
W2W product pitch (rubric below)	40%	No criterion at Level 1

Pitch score = total of the six criteria (each 1–4) ÷ 24 × 100. The website checks the quest score, lab badges and a complete pitch form; the facilitator verifies the pitch with this rubric and signs the portfolio sheet.

W2W Product Pitch rubric

Criterion	1 Beginning	2 Developing	3 Proficient	4 Exemplary
Problem and waste stream	Waste stream unclear; no link to a real problem.	Waste stream named; problem described in general terms.	Clear waste stream and local problem, with an estimate of kg diverted.	As Proficient, plus sourced data and a realistic plan to collect the waste.

Criterion	1 Beginning	2 Developing	3 Proficient	4 Exemplary
Product and process	Product unclear or not made.	Product made; steps incomplete.	Working product; steps clear and repeatable; science explained.	As Proficient, plus tested and improved, with evidence (e.g. a second version).
Safety and environmental impact	Hazards not identified.	Some hazards listed; controls incomplete.	Hazards and controls complete; no new pollution created.	As Proficient, plus life-cycle thinking: energy, water and the product's end of life.
Value proposition and costing	No customer or cost.	Customer named; cost or price incomplete.	Unit cost, price and margin calculated; customer need explained.	As Proficient, plus scale-up or “what if costs rise” and comparison with an alternative product.
Social impact	Not considered.	General benefit stated.	Specific community benefit (jobs, awareness, use in schools) linked to an SDG.	As Proficient, plus a plan to involve the community and measure the impact.
Communication	Hard to follow.	Main points clear; over time or few visuals.	Clear, on time, answers questions well.	Persuasive and well structured; uses data and visuals; confident in Q&A.

Quality assurance notes

- MQA defines micro-credentials as certification of assessed knowledge, skills and competencies in a specific field, from accredited programme components or stand-alone courses (MQA, 2020).
- Stand-alone micro-credentials are quality-verified through an **MQA-appointed Quality Verification Centre (QVC)** using form SAMC-01. The minimum is **40 NLH (1 credit)**, in steps of 20 NLH; no more than

2 ILOs per credit; apply within 3 months before the course or cohort ends; verification lasts 3 years (MQA GGP QVSAMC 2023, as summarised in university guidelines).

- **What this means:** at 20 NLH, the Practitioner is planned as a non-credit Certificate of Completion. Who issues it and what approval it needs are still to be confirmed with the Faculty and UPM CADe-Lead. It is below the 40 NLH minimum for MQA quality verification, which is voluntary; any formal micro-credential is a separate, approval-dependent route.
- Before applying, document: market need; LOs aligned to the MQF level; assessment matched to the LOs (at least two methods); facilitator qualifications; and secure records so certificates cannot be forged.
- An official certificate usually also shows a serial number, learner ID, learning outcomes, language, total learning hours, assessment type, grade and an authorised signature. The WasteQuest certificate currently shows name, track, date, score and a verification code.

Next steps (through UPM)

1. Pilot the 20-NLH Practitioner with one adult cohort; collect quest scores, rubrics and feedback.
2. Confirm the issuer and approval for the completion certificate with the Faculty and UPM CADe-Lead.
3. Only if a formal micro-credential is wanted later: develop it through UPM's micro-credential process (CADe-Lead and the offering Faculty). MQA quality verification would need at least 40 NLH.
4. Only after approval, update certificates and this page to say so.

🦺 Safety

⚠️ **Adult supervision required.** Heat, sharp tools, fermentation gas and electricity are used in some labs. Read each lab page before the session and prepare materials in advance.

Lab rules checklist

- [ ] An adult supervises every lab. For heat or sharp-tool steps, one adult per group.
- [ ] Safety briefing before starting; each group names a safety officer.
- [ ] Tie back long hair; closed shoes; apron. Goggles for hot liquids and pH tests.
- [ ] Use only clean waste. No broken glass, needles, batteries, medicines or chemical containers.
- [ ] Gloves for food scraps, fish waste and compost. Wash hands with soap afterwards.
- [ ] Never taste or eat anything in a lab, including the products.
- [ ] Adults handle stoves, hot wax, hot water and irons. Never leave heat unattended. Never put water on a wax or oil fire: turn off the heat and cover it with a lid.
- [ ] Adults pre-cut bottles with craft knives; children use safety scissors.
- [ ] Never melt or burn plastic in a pot, over a flame or in oil. The only plastic heating allowed: an adult ironing PE bags between baking paper in a ventilated room (Fused-Plastic lab).
- [ ] No lye (sodium hydroxide) with children. Adult groups making soap follow chemical-safety rules.
- [ ] Ask about allergies (fish, coffee, fragrance, latex gloves) before the lab.
- [ ] Label every jar or bottle with contents, name and date.
- [ ] Electronics: low-voltage USB or battery only, kept away from water.
- [ ] First-aid kit ready; emergency number 999. Clean up and sort leftover waste into the right bins.

🚫 **Never use:** the 2024 student method of melting plastic in hot engine oil. It gives off toxic fumes and is a serious fire risk.

Risk table for the 18 labs

Risk ratings are our guide for a typical school session with the controls in place. Review them for your own space and group.

Lab	Hazard	Risk	Control	Supervision
🕯️ **Eco-Candles from Used Cooking Oil**	Hot wax or oil; flame	High	Adult melts wax in a double boiler, never over direct flame; learners add colour and fragrance away from the stove; no water on a wax fire (lid on); light candles only when set and supervised.	Adult does all heating
🐟 **Fish-Waste Pet Treats**	Raw fish (bacteria); knives; stove heat	Medium	Gloves; separate board; adult cuts and cooks thoroughly; wash hands and surfaces; keep chilled; check fish allergies.	Adult cuts and cooks
🎁 **Trash to Treasure: Soap, Tote & Toy Car**	Scissors and needles; hot iron and fumes if fusing plastic bags	Medium	Safety scissors; needle count before and after; adult irons between baking paper on a low setting in a ventilated room; stop if it smokes.	Adult irons
🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste**	Hot water for extraction; test liquids	Low	Adult pours hot water; use only food-safe liquids (vinegar, lime juice, baking-soda water, soapy water); never bleach, drain or oven cleaner; goggles; do not taste.	Adult pours hot water

Lab	Hazard	Risk	Control	Supervision
☕ **Kopi Wira Bau: Coffee-Ground Odour Absorbers**	Mould on damp grounds; oven heat	Low	Dry grounds fully (sun or adult-operated oven); throw away mouldy batches; use breathable bags.	Adult uses the oven
🌱 **Self-Watering Bottle Garden**	Cut bottle edges; standing water (Aedes mosquitoes); electricity in the sensor version	Medium	Adult cuts with a craft knife and tapes edges; cover the reservoir and check it weekly; low-voltage electronics only, kept dry.	Adult cuts; weekly check
🍊 **Eco-Enzyme**	Gas pressure from fermentation; not drinkable; ants	Medium	Plastic bottles only, never sealed glass; leave headspace; release gas daily in the first month; label “Not for drinking”; store out of reach; dilute before use and patch-test on skin.	Adult checks weekly for 3 months
🧱 **Eco-Bricks & Eco-Bags**	Sharp plastic pieces; packing stick	Low	Only clean, dry soft plastic (dirty plastic grows mould); use a blunt wooden stick; gloves optional.	Normal supervision

Lab	Hazard	Risk	Control	Supervision
🪱 **Composting: Bokashi & Worms**	Microbes and fungal spores; pests; garden tools	Medium	Gloves and handwashing; no meat, dairy or oily food (attracts pests); keep the bin covered; learners with asthma or weak immunity do not turn dry compost.	Adult present when turning
🌽 **Cornstarch Bioplastic**	Stove heat; hot sticky mixture can spatter	Medium	Adult heats on low and stirs with a long spoon; goggles; cool fully before shaping.	Adult does all heating
🪴 **Bottle Vertical Garden**	Cut bottle edges; heavy wet soil overhead; standing water (Aedes); money plant is toxic if eaten	Low	Adult cuts and tapes edges; hang on a strong frame away from walkways; empty trays and check weekly; wash hands; edible plants and money plant kept apart and labelled.	Adult cuts
🥬 **Bottle Hydroponics**	Nutrient salts (not for drinking); standing water (Aedes); cut edges, nails and hammer	Medium	Adult measures nutrients from the label; label bottles “Not for drinking”; seal gaps with tape and check for wrigglers weekly; adult cuts and hammers; spent solution poured on plants, not drains.	Adult handles nutrients and tools

Lab	Hazard	Risk	Control	Supervision
☂️ **Fused-Plastic Bags & Mini Umbrella**	Hot iron; plastic fumes; needles	High	Adult irons only, low-medium with no steam, baking paper on both sides, windows open; PE bags (2 or 4) only, never PVC (3) or foil-lined wrappers; stop if it smokes or smells; needle count.	Adult does all ironing
🛟 **Bubble-Wrap Float Ring (Model)**	Drowning risk at the water basin; model mistaken for a real float; plastic film (suffocation for toddlers)	Medium	Adult stays at the basin and empties it after; say clearly it is a model, never for swimming or floods; teach “Reach or throw, don't go” and call 999; keep plastic away from toddlers.	Adult at the water
🛌 **3-in-1 Bubble-Wrap Sleeping Bag**	Needles and sewing machine; plastic over the face (suffocation); plastic catches fire	Low	Needle count; adult or older pupils use the machine; label the finished bag: not for babies or toddlers, never over the face, keep away from flames and mosquito coils.	Normal supervision; adult on the machine

📝 Rubrics & reflection

For kids and teens doing a hands-on lab. Score each criterion 1–4 (total out of 12). Teens can also peer-assess another group.

Hands-on rubric

Criterion	1 Beginning	2 Developing	3 Proficient	4 Exemplary
Product	Not finished.	Finished with a lot of help; some steps followed.	Finished; all steps followed; works as planned.	Works well and shows a creative improvement.
Teamwork and safety	Works alone or off task.	Joins in when asked.	Does own role, shares, follows safety rules.	Helps others, includes everyone, leads safely.
Reflection	Says only what was made.	Says what was made and from which waste.	Explains the science or why it helps the environment.	Also suggests an improvement and a real-life use.

Kids' star self-check

I can...	Colour the stars
I made my product safely.	☆ ☆ ☆
I followed the steps.	☆ ☆ ☆
I helped my team.	☆ ☆ ☆
I can say what waste we used.	☆ ☆ ☆
I can say how it helps the Earth.	☆ ☆ ☆

📝 Reflection worksheet

Name / group / date

What waste did we use, and where would it normally go?

What did we make? Draw or describe it.

What science made it work? (e.g. microbes, acids and bases, heat)

What went well? What would we change next time?

How much waste could this save in a month? Show your working.

Who could use or buy it?

My zero-waste promise for this week:

🏫 Curriculum links

WasteQuest supports, but does not replace, the national curriculum. The links below are general; always check the current DSKP for your year and subject, because standards and chapter numbers are revised.

Level	Subject and topic	WasteQuest activities
Primary (KSSR)	Science, upper primary: microorganisms, and preservation and conservation (Pemeliharaan dan Pemuliharaan) in Year 6. Check your DSKP.	🌱 Compost Master 🪱 Composting: Bokashi & Worms 🗑️ Sort It Out!
Form 1 (KSSM)	Design and Technology (RBT): Ch 3 Design Process, Ch 4 Sketching, Ch 5 Technology Application.	🌱 Self-Watering Bottle Garden 🎁 Trash to Treasure: Soap, Tote & Toy Car 🎤 W2W pitch form
Form 2 (KSSM)	Science: Ch 2 Ecosystems; Ch 6 Acids and Alkalis.	🌱 Compost Master 🧪 Say No to Litmus! 🌈 Say No to Litmus: Rainbow pH from Kitchen Waste 🍊 Eco-Enzyme
Secondary (KSSM)	Geography: environment and sustainability topics (check the DSKP for your form).	📘 Learn 1 📘 Learn 6 👣 My Waste Footprint
All KSSM subjects	Cross-curricular elements (EMK): Environmental Sustainability, Entrepreneurship, Creativity and Innovation, Science and Technology.	💰 Trash-to-Cash Tycoon 🏟️ Class Battle
University and adults	Sustainability, engineering and society, ESG reporting (e.g. GRI 306: Waste), EPR, SDG 12.	📘 Learn 3 📘 Learn 4 📘 Learn 6 🎤 W2W pitch form

These are suggested links only. Please check them against the DSKP edition your school uses.

Running Class Battle

Class Battle is a live quiz for the whole class: the questions show on the projector and students answer on their phones. **Class Battle**

Steps

1. Open WasteQuest on the computer connected to the projector and go to Class Battle.
2. Choose **Host a class**. Pick the level (kids, teens, adults), topic, number of questions and seconds per question.
3. Students scan the QR code, or open WasteQuest → Class → Join a class and type the 5-character code. No app or login; nicknames only.
4. When everyone is in, press **Start quiz**. Faster correct answers earn more points; the leaderboard shows after each question.
5. At the end, download the results (CSV). The "% correct" per question shows which topics to teach again.

Needs internet. Live mode links phones to the projector through a free online service (PeerJS). It does not work when WasteQuest is opened from a file or localhost: use the online site. School Wi-Fi may block it; a phone hotspot usually works.

No phones or no internet? Use **Team mode**: 2–4 teams play on the projector only, offline. After each question, tap every team that got it right (+100 points).

Teaching tips

- Warm-up: 5 questions at the start to find out what the class already knows.
- Exit ticket: 5 questions at the end; reteach the question with the lowest % correct next lesson.
- Remind students to use kind nicknames; remove any unsuitable name from the lobby.
- Pause after each question to explain the answer: the discussion is where most learning happens.

🤝 Who can use it

Who can use WasteQuest

User	Typical use	Suggested plan
Primary and secondary schools	Science, RBT and Eco-Club lessons; school recycling drives	60-minute lesson; Eco-Club
Universities and TVET	Sustainability and engineering-and-society courses; student outreach projects	Workshop; Practitioner
Government agencies and local authorities	Public awareness on separation at source and recycling	Workshop; community package
NGOs and community learning centres	Community workshops and W2W enterprise ideas	Workshop; Eco-Club
Training providers	Sustainability and ESG courses	Adult half-day; Practitioner
Companies and green-technology organisations	Staff ESG training; CSR programmes with schools	Adult half-day; Practitioner

Packages

Package	What it includes
📱 Interactive digital learning module	Free bilingual website for self-directed and classroom learning: Learn chapters, games, simulations, labs, Class Battle, certificates.
🏫 School environmental education toolkit	60-minute lesson, Eco-Club plan, rubrics, reflection worksheet, safety checklist.
🎒 Sustainability educational package	Workshop with facilitator, lab materials and certificates.
💼 ESG training module	Adult half-day and the 20-hour Practitioner Certificate of Completion (non-credit).
🏘️ Community sustainability learning package	Workshop plus community labs (compost, eco-enzyme, ecobricks).

The website is free. Facilitated workshops, kits and adult courses are quoted separately, based on group size, activities, materials, venue and travel; duration, certificate and fee are confirmed in the quotation. Bookings are not open yet.

🙏 Credits & citation

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6. Appendix: Question Bank & Answers

1. [KT] Which SWCorp recycling bin is for paper?
A. Blue bin B. Orange bin C. Brown bin

2. [KT] Where does a clean, empty plastic bottle go?
A. Brown bin **B. Orange bin** C. Blue bin

3. [KT] Which bin is for a glass jam jar?
A. Orange bin B. Blue bin **C. Brown bin**

4. [KT] An empty aluminium drink can goes into the...
A. Orange bin B. Brown bin C. Blue bin

5. [K] What should you do to a bottle before recycling it?
A. Fill it with sand **B. Empty it and rinse it** C. Break it into pieces

6. [KT] Can used tissue paper go into the blue paper bin?
A. No: it is dirty and its fibres are too short B. Yes: all paper can be recycled

7. [KT] Where should old batteries go?
A. A battery or e-waste collection point B. The normal rubbish bin at home C. The compost heap in the garden

8. [KT] What should you do with expired medicine?
A. Flush it down the toilet **B. Return it to a pharmacy or clinic** C. Put it in the compost

9. [KTA] Why is it important to separate waste at home?
A. It makes the rubbish heavier B. It means rubbish lorries do not need to come **C. It keeps recyclables clean so they can really be recycled**

10. [KT] Where does a banana peel belong?
A. Compost B. Blue paper bin C. Brown glass bin

11. [KT] A mirror breaks. What is the right way to throw it away?
A. Put it in the brown bin with glass bottles **B. Wrap it safely and put it in general waste** C. Put it in the compost

12. [TA] Why must lithium-ion batteries (phones, power banks) never go into general waste?
A. They are too light for lorries to collect B. They melt harmlessly when it rains **C. They can catch fire when crushed in lorries**

13. [TA] Fluorescent lamps need special disposal because they contain...
A. Sugar B. Salt **C. Mercury** D. Iron

14. [KT] Old phones contain precious metals such as...
A. Sand and salt **B. Gold and copper** C. Wood and paper

15. [TA] Which of these is household hazardous waste?
A. Paint thinner and pesticide containers B. Banana peels and other fruit scraps C. Old newspapers and magazines D. Empty, rinsed glass jars

16. [A] What usually happens to a load of recyclables that is heavily contaminated with food?
A. Much of it is rejected and ends up in landfill B. It is recycled into higher-value products C. It turns into compost on its own

17. [TA] What does “urban mining” mean?
A. Digging new mines underneath big cities **B. Recovering metals from e-waste instead of mining ore** C. Burying e-waste in city landfills to store it

18. [TA] Why is burning electrical cables at home to get the copper dangerous?
A. It is only dangerous because it is slow B. Copper explodes when heated **C. The burning plastic gives off toxic fumes that harm health**

19. [K] Which of these is e-waste?
A. A broken phone charger B. An apple core C. A cardboard box

20. [KTA] What does “Waste-to-Wealth” mean?
A. Burning all rubbish **B. Turning waste into useful, valuable things** C. Hiding rubbish underground

21. [TA] Which of these is an ENVIRONMENTAL benefit of Waste-to-Wealth?
A. A small business earns income B. Neighbours learn new skills together **C. Less waste goes to landfill**

22. [TA] Which of these is an ECONOMIC benefit of Waste-to-Wealth?
A. Selling waste-based products creates income and jobs B. Less methane escapes from landfill sites into the air C. Neighbours learn new skills and work together

23. [TA] Which of these is a SOCIAL benefit of Waste-to-Wealth?
A. Factories pay less for the raw materials they buy **B. Community workshops build skills and bring people together** C. Less leachate seeps from landfills into the soil

24. [K] Which of these is a Waste-to-Wealth product?
A. A brand-new plastic bottle from a factory B. Rubbish burnt in the backyard **C. A candle made from used cooking oil**

25. [KT] Dried used coffee grounds can be made into...
A. An odour absorber B. Drinking water C. Glass

26. [A] Turning old jeans into a bag worth more than the jeans is called...
A. Downcycling B. Landfilling C. Incineration **D. Upcycling**

27. [A] Which of these is an example of downcycling?
A. Turning an old T-shirt into a tote bag B. Repairing a broken chair **C. Turning white office paper into egg cartons**

28. [KT] Fish waste such as heads and bones can be turned into...
A. Pet food or fertiliser B. Glass bottles or jars C. Batteries or wires

29. [KTA] What can an old T-shirt become?
A. Petrol **B. A tote bag** C. A glass jar

30. [KTA] A “linear economy” works like this:
A. Make → use → reuse forever B. Grow → eat → compost **C. Take → make → throw away**

31. [KTA] In a circular economy, materials are…
A. Kept in use for as long as possible B. Used once, then thrown away C. Burnt for heat straight away

32. [A] Which is NOT one of the three circular-economy principles of the Ellen MacArthur Foundation?
A. Eliminate waste and pollution B. Circulate products and materials C. Regenerate nature **D. Maximise sales of single-use products**

33. [K] Which is BEST for the planet?
A. Throwing it in the rubbish bin quickly B. Burning it in the back garden **C. Not making the waste in the first place**

34. [TA] In the waste hierarchy, which option is MOST preferred?
A. Recycling (making new items from old) **B. Prevention (reducing waste at source)** C. Energy recovery (burning for power) D. Landfill (burying waste)

35. [TA] In the waste hierarchy, which option is LEAST preferred?
A. Reuse (using items again) B. Recycling (making new items)
C. Landfill (burying waste) D. Reduce (making less waste)

36. [A] Where does energy recovery (waste-to-energy) sit in the waste hierarchy?
A. Above reduce and reuse B. At the very top **C. Below recycling, but above landfill**

37. [KT] Saying "no thanks" to a plastic straw is an example of…
A. Refuse B. Recycle C. Landfill

38. [K] Using an old jam jar as a pencil holder means you…
A. Throw it away (dispose of it) **B. Give it a new job (reuse it)** C. Melt it down (recycle it)

39. [T] Turning a jam jar into a pencil holder is best described as…
A. Recycle B. Refuse **C. Repurpose**

40. [TA] In the 5R list (Refuse, Reduce, Reuse, Repurpose, Recycle), which should be the LAST choice?
A. Recycle B. Refuse C. Reduce

41. [KT] Fixing a broken toy instead of buying a new one…
A. Makes more rubbish for the bin **B. Keeps it in use and saves resources** C. Uses up more of the planet's resources

42. [A] Which business model is the most circular?
A. Selling cheaper single-use items in bigger volumes B. Adding more packaging to look premium **C. Leasing products, then taking them back to refurbish (product-as-a-service)**

43. [A] "Design for disassembly" supports a circular economy because…
A. Products are easy to repair and to separate for recycling
B. Products wear out faster, so people buy new ones C. Parts are glued firmly so nothing comes loose

44. [A] A life cycle assessment (LCA) measures…
A. Only the factory's energy bill in one year **B. Environmental impacts over a product's whole life** C. Only how long the product lasts before breaking

45. [A] A refill shop where customers bring back empty containers is an example of…
A. Recycling in a linear business model B. Energy recovery through incineration **C. Reuse in a circular business model**

46. [K] Which one is a "green" for the compost bin?
A. Dry leaves **B. Vegetable scraps** C. A plastic spoon

47. [K] Which one is a "brown" for the compost bin?
A. Fresh grass B. A glass bottle **C. Dry leaves**

48. [TA] In composting, "greens" are mainly a source of…
A. Nitrogen B. Carbon only C. Plastic

49. [TA] What is a good starting carbon-to-nitrogen (C:N) ratio for compost?
A. About 25–30 : 1 B. About 1 : 1 C. About 5 : 1 D. About 100 : 1

50. [TA] How wet should a compost pile be?
A. Completely dry B. Soaking, with water pooling at the bottom **C. Like a wrung-out sponge (about 40–60% moisture)**

51. [TA] A compost pile smells of rotten eggs. What is the most likely cause and fix?
A. Too wet, not enough air: add browns and turn it B. Too dry, too much air: add water and sand C. Too cold: cover it tightly with a plastic sheet

52. [TA] Why is a "hot" compost pile (about 55–65 °C) useful?
A. The heat melts plastic into the compost **B. The heat kills many pathogens and weed seeds** C. It turns compost into soil in one hour

53. [KT] Who does most of the work in a compost pile?
A. Magnets hidden in the soil B. Only the hot sun shining on it
C. Tiny microbes like bacteria and fungi

54. [KTA] Which should NOT go into a simple home compost bin?
A. Meat, bones and oily food B. Vegetable peels and tea leaves C. Dry leaves and small twigs

55. [TA] What is bokashi?
A. Burning dried food waste in a small stove for energy **B. Fermenting food waste in a sealed bucket with microbe bran**
C. Composting food waste in an open pile using only worms

56. [A] What must be done with the fermented material from a bokashi bucket?
A. Use it straight away as potting soil for seedlings B. Send it to the blue recycling bin **C. Bury it in soil or add it to a compost heap to finish breaking down**

57. [K] Which animal helps make vermicompost?
A. Earthworms B. Ants C. Cats

58. [TA] Vermicomposting usually uses…
A. Garden snails and slugs **B. Composting worms (red wigglers)** C. Soil ants and termites

59. [TA] Why do we turn (mix) a compost pile?
A. To cool it down to freezing B. To remove all the nitrogen **C. To add oxygen for the microbes**

60. [A] Your compost has lots of food scraps (C:N about 15:1). To move towards 30:1 you should add...
A. Browns such as dry leaves or shredded cardboard B. Greens such as fresh grass clippings C. More fruit peels and vegetable scraps

61. [KT] Finished compost looks and smells...
A. Bright and shiny **B. Dark, crumbly and earthy** C. Like rotten eggs

62. [K] Eco-enzyme is made from fruit peels, water and...
A. Salt or baking soda **B. Brown sugar or molasses** C. Cooking oil or flour

63. [TA] The usual eco-enzyme recipe ratio (by weight) of sugar : peels : water is...
A. 1 : 3 : 10 B. 3 : 1 : 10 C. 10 : 3 : 1 D. 1 : 1 : 1

64. [KTA] How long does eco-enzyme usually ferment?
A. About 3 months B. About 3 hours C. About 3 days

65. [KT] Why do we open the eco-enzyme lid a little in the first weeks (with an adult)?
A. To let flies in to help it ferment **B. To let out gas so it does not burst** C. To add sand to make it heavier

66. [TA] Which container is best for fermenting eco-enzyme?
A. A glass bottle filled to the brim and sealed tight B. An open bowl **C. A plastic container with space left at the top**

67. [TA] Which statement about eco-enzyme is honest?
A. A mild fermented cleaner; cure and river-cleaning claims are unproven B. A proven medicine that cures many diseases when you drink it C. A powerful treatment that instantly cleans any polluted river

68. [T] Finished eco-enzyme is acidic (low pH) mainly because...
A. The brown sugar itself is a strong acid **B. Fermentation makes organic acids such as acetic acid** C. The peels release metals into the water

69. [KT] Which waste is good for making eco-enzyme?
A. Meat and fish bones B. Plastic food wrappers **C. Fruit and vegetable peels**

70. [A] Why should large amounts of eco-enzyme NOT be poured into rivers to "clean" them?
A. Its sugars use up the oxygen that fish need B. It makes river water too cold for fish C. It turns the river water into plastic

71. [TA] Using the 1 : 3 : 10 ratio, how much peel and water go with 300 g of brown sugar?
A. 300 g peels and 300 g water **B. 900 g peels and 3 kg (about 3 L) water** C. 3 kg peels and 900 g water

72. [KT] A pH of 7 means the liquid is...
A. Neutral B. Very acidic C. Very alkaline

73. [KT] Red (purple) cabbage juice turns pink-red when you add...
A. Soapy water **B. Vinegar** C. Plain water

74. [KT] Blue bunga telang (butterfly pea) tea turns purple-pink when you add...
A. Baking soda B. Salt **C. Lime juice**

75. [K] Red cabbage juice in baking soda water turns...
A. Blue-green B. Bright red C. Stays exactly the same

76. [T] Which natural pigment makes red cabbage and butterfly pea change colour with pH?
A. Chlorophyll B. Starch **C. Anthocyanins** D. Protein

77. [T] Turmeric (kunyit) solution turns from yellow to reddish-brown in...
A. Acidic solutions such as vinegar B. Neutral liquids such as pure water **C. Alkaline solutions such as soapy water**

78. [KT] Which of these is acidic?
A. Lemon juice B. Soap C. Baking soda solution

79. [KT] Blue litmus paper turns red in...
A. An alkali **B. An acid** C. Pure water

80. [T] Compared with a solution at pH 5, a solution at pH 3 is...
A. 2 times more acidic B. Less acidic **C. 100 times more acidic**

81. [TA] Why make pH indicators from kitchen waste (e.g. cabbage leaves, flower petals) in class?
A. They are cheap, safe and reuse waste: a W2W science tool B. They are more precise than a pH meter C. They only work at night

82. [TA] Resin code 1 (♳) on a drink bottle stands for...
A. PVC (polyvinyl chloride) B. PS (polystyrene) **C. PET (polyethylene terephthalate)** D. PP (polypropylene)

83. [TA] Which resin code is HDPE (e.g. shampoo and detergent bottles)?
A. 4 B. 6 C. 7 **D. 2**

84. [TA] Resin code 4 is LDPE. Which product is typically LDPE?
A. Polystyrene food boxes **B. Plastic bags and cling film** C. PVC water pipes

85. [TA] Polystyrene (foam food boxes) has which resin code?
A. 1 **B. 6** C. 2 D. 5

86. [TA] Resin code 5 (PP) is commonly used for...
A. Microwave-safe tubs and bottle caps B. Clear fizzy-drink and water bottles C. Foam takeaway boxes and cups

87. [A] Resin code 7 means...
A. The easiest plastic of all to recycle **B. "Other" plastics or mixtures, hard to recycle** C. Clear PET drink bottles

88. [KTA] What are microplastics?
A. Extra-strong, thin plastic film B. Plastic made by tiny microbes **C. Plastic pieces smaller than 5 mm**

89. [TA] Which is a major source of microplastics in wastewater?
A. Polyester clothes shedding fibres in the wash B. Glass bottles broken in recycling bins C. Paper towels used to wipe kitchens

90. [KT] How long can a plastic bottle last in nature?
A. About one week, then it melts away **B. Hundreds of years, breaking into tiny bits** C. A few months, like a banana peel

91. [KTA] What is the best way to cut plastic waste?
A. Use more plastic straws but recycle them B. Burn plastic at home to make it disappear **C. Refuse single-use plastic; bring a reusable bag**

92. [TA] Why is open burning of plastic waste harmful?
A. It releases toxic smoke (e.g. dioxins) and is illegal B. It is only harmful if you burn it in the daytime C. It makes the ash too heavy for rubbish lorries

93. [TA] Which statement about "bio-based" plastics is correct?
A. All bio-based plastics dissolve in the sea within days **B. Bio-based means made from plants; it is not always biodegradable** C. Bio-based plastics are always made from petroleum

94. [TA] PLA "compostable" cups break down properly mainly in…
A. The cool, salty ocean B. A cool backyard compost heap **C. Hot industrial composting facilities**

95. [KT] Which ingredients make a simple cornstarch bioplastic?
A. Cornstarch, water, vinegar and glycerol B. Cornstarch, sand, cement and salt C. Flour, petrol, salt and sugar

96. [T] In cornstarch bioplastic, what does glycerol do?
A. It makes the film hard and brittle **B. It is a plasticiser that makes the film flexible** C. It is a dye that gives the film colour

97. [TA] Why are foil-lined snack packets hard to recycle?
A. Their bright inks poison recycling machines B. They are made of thin sheets of glass **C. Plastic and foil layers are glued together**

98. [KT] Sea turtles sometimes eat plastic bags because…
A. Floating bags look like jellyfish, their food B. Plastic tastes sweet C. Turtles collect plastic for nests

99. [A] Why can "compostable" plastic items cause problems in normal plastic recycling?
A. They make recycled plastic too strong to mould **B. They contaminate the stream and weaken recycled plastic** C. They turn the recycling plant into a compost site

100. [KTA] Why shouldn't used cooking oil be poured down the sink?
A. It cleans the pipes and makes them shine **B. It hardens, blocks pipes and pollutes rivers** C. It turns into clean water in the drain

101. [KTA] Used cooking oil can be turned into…
A. Drinking water or juice B. Glass jars or bottles **C. Candles, soap or biodiesel**

102. [KT] What is the first step before making a candle from used cooking oil?
A. Filter out the food bits B. Add cold water to it C. Freeze the oil solid

103. [K] A lab uses hot wax or a stove. Who should handle the hot parts?
A. The youngest child, so they can learn **B. An adult; children help with the cool steps** C. Anyone at all, with nobody watching

104. [TA] Making soap from oil uses lye (sodium hydroxide, NaOH). What is the key safety rule?
A. Taste a tiny drop to check its strength B. Stir it quickly with bare hands to save time **C. Adult supervision, gloves and goggles: lye burns skin**

105. [TA] The reaction of oil or fat with sodium hydroxide to make soap is called…
A. Photosynthesis **B. Saponification** C. Fermentation D. Combustion

106. [TA] Biodiesel is made from used cooking oil by transesterification. The main by-product is…
A. Petrol **B. Glycerol** C. Salt water

107. [TA] A candle recipe uses oil : wax = 1 : 2. You have 50 g of filtered oil. How much wax do you need?
A. 25 g B. 50 g **C. 100 g**

108. [KT] How does a self-watering plant pot made from a plastic bottle keep the soil moist?
A. A cloth wick pulls water up from below B. The plastic slowly turns into water C. Sunlight pumps water up through the cap

109. [TA] What is an eco-brick?
A. A brick made by melting glass bottles in a kiln **B. A bottle packed hard with clean, dry plastic, used for building** C. A block of dried compost pressed into a mould

110. [KT] Why must plastic stuffed into an eco-brick be clean and dry?
A. Wet plastic packs in tighter and stronger B. Clean plastic makes the bottle lighter **C. Leftover food would rot, smell and grow germs**

111. [A] Why is collecting used cooking oil for biodiesel valued in a circular economy?
A. It turns polluting waste oil into a diesel substitute B. It makes fresh cooking oil cheaper to produce C. It sends more waste oil safely to landfill

112. [T] Why should a home-made pet food from fish waste be cooked thoroughly (with an adult)?
A. Cooking makes the food a nicer colour **B. Heat kills harmful germs in raw fish waste** C. Raw fish waste is too light to measure

113. [KT] In a bottle vertical garden, why do the bottles need small holes underneath?
A. So insects can get in to eat the plants B. So the bottle is lighter to carry **C. So extra water drains and the roots get air**

114. [TA] In the Kratky bottle hydroponics method, why should you NOT keep refilling the bottle to the top?
A. The upper roots need an air gap to get oxygen B. Plants can only drink from a half-empty bottle C. Nutrients only dissolve in a small amount of water

115. [TA] Which plastic must NEVER be ironed or heated to make fused plastic fabric?
A. LDPE carrier bags (code 4) **B. PVC (resin code 3)** C. HDPE carrier bags (code 2)

116. [KTA] A bubble-wrap float ring holds up 2 kg in a basin. What is it safe to use it for?
A. Rescuing a child in a flood B. Swimming in the sea **C. Only as a science model, never for swimming or floods**

117. [KT] Why does a bubble-wrap mat keep you warmer on a cold floor?
A. The trapped air in the bubbles is bad at carrying heat B. Plastic makes its own heat C. The bubbles reflect all the cold away

118. [K] About how much rubbish does each person in Malaysia throw away every day?
A. About 1 kg: like a 1-litre bottle of water B. About 10 g: like a pencil C. About 100 kg: like a fridge

119. [TA] How much solid waste did Malaysia generate per day in 2024 (SWCorp)?
A. About 3,900 tonnes **B. About 39,000 tonnes** C. About 390,000 tonnes

120. [TA] How much waste does the average Malaysian generate per day?
A. About 0.17 kg B. About 11.7 kg **C. About 1.17 kg**

121. [KTA] What is the BIGGEST part of household rubbish in Malaysia?
A. Food waste B. Glass C. Metal

122. [TA] What share of Malaysian household waste is food waste?
A. About 3% **B. About 31%** C. About 80%

123. [TA] After food, which is the second-largest part of Malaysian household waste?
A. Glass (about 3%) B. Textiles (about 2%) **C. Plastic (about 22%)**

124. [TA] How did Malaysia's daily solid waste change from 2005 to 2024?
A. It roughly doubled B. It roughly halved C. It stayed about the same

125. [TA] What is Act 672?
A. Environmental Quality Act 1974 **B. Solid Waste and Public Cleansing Management Act 2007** C. Street, Drainage and Building Act 1974

126. [TA] In which of these places is household separation at source mandatory under Act 672?
A. Sabah B. Pulau Pinang C. Sarawak **D. Johor**

127. [TA] Which item makes up a surprisingly large 8.2% of Malaysian household waste?
A. Disposable diapers B. Face masks C. Rubber

128. [TA] A family throws away 10 kg of household waste a week. Using the national average, about how much is food?
A. About 1 kg **B. About 3 kg** C. About 6 kg

129. [A] A town of 10,000 people generates waste at the national average of 1.17 kg/person/day. How much per day?
A. 1.17 tonnes B. 117 tonnes **C. 11.7 tonnes**

130. [A] At 1.17 kg per day, about how much waste does one person generate in a year?
A. About 430 kg B. About 43 kg C. About 4.3 tonnes

131. [TA] About what share of Malaysian household waste is hazardous household waste (batteries, chemicals and similar)?
A. About 40% **B. About 4%** C. About 0.04%

132. [KT] In Malaysia, what colour is the SWCorp recycling bin for glass?
A. Blue B. Orange **C. Brown**

133. [KTA] SDG 12 is about...
A. Responsible consumption and production B. Clean water and sanitation C. Industry, innovation and infrastructure

134. [KTA] SDG 13 is...
A. Quality education **B. Climate action** C. Clean water and sanitation

135. [KTA] SDG 15 is...
A. Life below water B. Affordable and clean energy **C. Life on land**

136. [KTA] How many Sustainable Development Goals (SDGs) are there?
A. 17 B. 10 C. 25

137. [TA] By which year are the SDGs meant to be achieved?
A. 2050 **B. 2030** C. 2025

138. [A] SDG target 12.3 aims to...
A. Ban all single-use plastics worldwide by 2030 B. Double recycling rates for all household waste by 2030 **C. Halve per-capita food waste at retail and consumer level by 2030**

139. [TA] Which SDG most directly covers plastic pollution in the ocean?
A. SDG 14 Life below water B. SDG 4 Quality education C. SDG 16 Peace and justice

140. [KT] A school compost garden that makes the soil healthy supports which SDG?
A. SDG 1 No poverty **B. SDG 15 Life on land** C. SDG 16 Peace and justice

141. [A] A W2W social enterprise that pays fair wages to local women making soap from used oil links most directly to...
A. SDG 14 Life below water B. SDG 6 Clean water and sanitation **C. SDG 8 Decent work and economic growth**

142. [TA] Composting food waste instead of landfilling it cuts methane. Which SDG does this support most directly?
A. SDG 13 Climate action B. SDG 4 Quality education C. SDG 16 Peace and justice

143. [KTA] Food rotting in a landfill makes which greenhouse gas?
A. Oxygen **B. Methane** C. Helium

144. [TA] Over 100 years, one tonne of methane warms the climate about how much more than one tonne of CO_2?
A. The same B. Half as much **C. About 28 times**

145. [TA] Why do landfills produce methane?
A. Organic waste rots without oxygen (anaerobic) B. Sunlight reacts with the plastic on top C. Rainwater dissolves metals in the rubbish

146. [A] What does landfill gas capture do?
A. Pumps fresh air in so the rubbish dries out **B. Captures methane and burns it, often for electricity** C. Releases the gas faster so the site settles sooner

147. [KT] What does a greenhouse gas do?
A. Blocks sunlight, making the Earth colder B. Turns leaves purple, making plants sick **C. Traps heat, making the Earth warmer**

148. [TA] Making a can from recycled aluminium saves up to about how much energy compared with new aluminium from ore?
A. About 95% B. About 5% C. None

149. [K] Which choice saves the most energy?
A. Buying a new bottle every day **B. Refilling the same water bottle every day** C. Throwing bottles in the drain

150. [TA] Biogas from the anaerobic digestion of food waste is mainly...
A. Oxygen and nitrogen B. Hydrogen and helium **C. Methane and carbon dioxide**

151. [A] Which is a real concern with waste-to-energy incineration?
A. Long waste-supply contracts can discourage recycling, and emissions need strict control B. It leaves no ash and releases no emissions, so it needs no pollution controls C. It sits at the top of the waste hierarchy, above reduce and reuse

152. [A] A product's "carbon footprint" is...
A. How much carbon (graphite) is inside it **B. Total greenhouse gases it causes, in CO_2-equivalent** C. The weight of its packaging in kilograms

153. [A] ESG stands for...
A. Energy, Sales and Growth B. Eco, Safety and Green **C. Environmental, Social and Governance**

154. [A] Which ESG pillar covers community programmes and worker safety?
A. Social B. Environmental C. Governance

155. [A] Which is an example of the GOVERNANCE pillar of ESG?
A. Measuring carbon emissions **B. Board oversight, business ethics and anti-corruption policies** C. Running community recycling workshops

156. [A] What is greenwashing?
A. Washing products with plant-based green soap B. Painting factories green so they blend with nature **C. Misleading claims that something is greener than it is**

157. [A] Which is a red flag for greenwashing?
A. Vague words such as "eco-friendly" or "natural" with no evidence B. A clear percentage of recycled content with proof C. An independent third-party certification

158. [A] A brand labels its pouch "100% recyclable", but no facility in Malaysia accepts that material. This claim is...
A. Honest: the material can be recycled somewhere **B. Misleading: recyclable in theory only** C. An example of extended producer responsibility

159. [A] What does Extended Producer Responsibility (EPR) mean?
A. Consumers pay all waste collection and recycling costs B. The government bans packaging on all imported products **C. Producers pay for and manage their products after use**

160. [A] Why does EPR encourage better product design?
A. Paying end-of-life costs rewards less material and easier recycling B. It requires all packaging to be heavier so it lasts longer C. It shifts all recycling costs back to households and councils

161. [A] Bursa Malaysia requires listed companies to publish...
A. Their product recipes **B. A sustainability statement in their annual report** C. Their employees' home addresses

162. [A] Emissions from a company's waste sent to landfill are reported under...
A. Scope 1 (own fuel burning) B. Scope 2 (purchased electricity) **C. Scope 3 (value chain)**

163. [A] The "triple bottom line" measures a business by...
A. People, planet and profit B. Price, place and promotion C. Plastic, paper and plates

164. [TA] A used-oil candle costs RM3.00 to make and sells for RM5.00. What is the profit per candle?
A. RM8.00 **B. RM2.00** C. RM5.00

165. [A] Same candle: cost RM3.00, price RM5.00. What is the profit MARGIN (profit ÷ price)?
A. 67% B. 60% C. 2% **D. 40%**

166. [A] Same candle: cost RM3.00, price RM5.00. What is the MARKUP on cost (profit ÷ cost)?
A. About 67% B. About 40% C. About 167%

167. [A] Moulds and tools cost RM200 (fixed). Each candle costs RM3 to make and sells for RM5. How many must you sell to break even?
A. 40 **B. 100** C. 67 D. 200

168. [TA] A bar of used-oil soap needs RM1.20 of materials and RM0.30 of packaging. It sells for RM4.00. Profit per bar?
A. RM2.80 B. RM1.50 **C. RM2.50** D. RM4.00

169. [A] You sell 200 soap bars a month at RM2.50 profit each. Monthly profit?
A. RM250 B. RM50 C. RM5,000 **D. RM500**

170. [A] A batch of 20 candles uses RM30 of materials and RM10 of electricity and packaging. What is the unit cost?
A. RM2.00 B. RM1.50 C. RM0.50 D. RM40.00

171. [A] Each candle uses 50 g of used cooking oil. Making 400 candles a month diverts how much oil from drains?
A. 2 kg B. 200 kg **C. 20 kg**

172. [A] Which evidence best supports an honest environmental claim for a W2W product?
A. Records of kilograms of waste diverted each month B. A bold "Saves the planet!" slogan on the label C. Green packaging with leaf and globe pictures

173. [A] Why can waste streams such as used oil or coffee grounds give a W2W business a cost advantage?
A. Waste never needs cleaning, sorting or testing **B. They are low-cost inputs and suppliers avoid disposal fees** C. Products made from waste are exempt from safety rules

174. [TA] Your eco-soap sells at RM4.00 but costs RM4.50 to make. What does this mean?
A. You make RM0.50 profit per bar: keep going B. You break even: costs and sales match **C. You lose RM0.50 per bar: cut costs or raise the price**

175. [K] Which of these can a recycling centre pay you money for?
A. Clean aluminium drink cans B. Used tissue paper C. Leftover rice

176. [K] Used cooking oil can be collected and sold. What should you keep it in?
A. Down the kitchen sink **B. A closed bottle, after it cools** C. An open bowl outside

K = kids · T = teens · A = adults

177. [K] Your family refills the same water bottle every day. What happens?
A. You spend more money B. You make more rubbish **C. You save money and make less rubbish**

178. [K] A school sells bags of compost made from garden and food scraps. This is an example of...
A. Waste to wealth B. Throwing money away C. Open burning

179. [K] You make 5 bookmarks from old cards and sell each for RM1. How much do you get?
A. RM1 **B. RM5** C. RM10

180. [K] Why should recyclables be clean before you take them to a recycling centre?
A. Clean items are heavier B. Dirt makes them shiny **C. Dirty items are worth less or get rejected**